الشركة السعودية للكهرباء
Saudi Electricity Company

TES-S-101

Revision No. 0

SAFETY AND SECURITY STANDARDS

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
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Date of Approval: December 15, 2007		

TRANSMISSION ENGINEERING STANDARD

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
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TES-S-101.01

Revision No. 0

SAFETY & SECURITY – INTRODUCTION

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1. INTRODUCTION

1.1 The TES-S-101 series of the Saudi Electricity Company Engineering Standards have been established to cover Safety and Security Directives (SSD) received from the Saudi Arabian Government (SAG) High Commission for Industrial Security (HCIS).

2. PURPOSE

2.1 The purpose of the TES-S-101 Standards is to incorporate HCIS Safety and Security directives in the engineering design of vital SEC facilities to insure the safe operations of, or minimize the damage to these vital facilities. These TES-S-101 Standards shall apply only to facilities which have been designated vital facilities by the High Commission for Industrial Security or as specified by the facility Design Based Memorandum.

2.1.1 Vital is further defined by the HCIS as

Quote

"Any piece of equipment vulnerable to sabotage, in or connected with a vital installation, where destruction or damage would result in unacceptable interruption of the installations services."

Unquote.

3. PROCEDURE

3.1 The proposed design for all SEC facilities which require SSD compliance shall be presented to the HCIS for approval by the Coordinator of SEC Technical Security. Four (4) copies of the Design Basis Memorandum drawing shall be made available for the HCIS review. The HCIS will advise if further approval of design drawings shall be required. The design effort shall not proceed until approval in writing of the drawings is received from HCIS.

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No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
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TES-S-101.02

Revision No. 0

SAFETY & SECURITY – SECURITY FENCE

PROPRIETARY INFORMATION

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1. SCOPE

This Standard refers to security fence and is adopted verbatim from SAG SSD/1.

Quote

1. Where chain-link fencing is used for security purposes the following specific requirements are to be met:

   a. The chain-link fencing shall be not less than 2.4m in height measured from the ground line (graded) to the top of the chain-link fencing at the posts.

   b. The total height of the fence including any cranked extension, shall be not less than 2.9m in vertical height measured from the ground line (graded) to the top line of the barbed wire attached to the cracked extension.

2. PVC coated chain-link fencing is to be used. The bond between the vinyl coating and the metal surface shall be not less than the cohesive strength of the vinyl. The colour of the PVC shall be dark green or black.

3. Barbed wire shall consist of three rows of 2 strand 12.5 SWG galvanized 4-point thick-set barbed wire, secured to the posts with 14.5 SWG galvanized wire loops.

4. The top lines of the barbed wire shall be fitted with droppers so that they cannot be bunched together.

5. As far as possible the fence is to run in straight lines. Where changes of direction are required the angle should be not less than 130º.

6. The fence to be sunk into a concrete channel 150mm x 150mm to prevent burrowing. The channel to be at minimum 50mm below the ground line (graded).

7. Where the fence is erected on a coast, the minimum distance from the fence to the high water mark shall be not less than 60m.

8. A clear zone is to be created on either side of the fence to a minimum depth of 15m.

9. The following specifications will apply for all wiring and concrete used in the construction of fencing:

   a. The chain link shall be woven from galvanized wire with an outside diameter of 4.75mm including PVC coating with a core of 3.5mm to a mesh size of 50mm.

   b. The top of the fencing shall be protected by the addition of a single line of barbed wire. It shall be secured to the cleat at a straining post and to be chain-link where knuckled at 150mm center with tying wire or suitable clips.

c. Line Wire shall have a core of 3.55mm diameter and an outside diameter of 4.75mm.

d. Stirrup Wire for securing the line wires to the concrete intermediate posts shall have a core of 2.5mm and an outside diameter of 3.55mm.

e. Tying Wire for securing the chain-link fencing to the line wires shall have a core of 1.4mm and an outside diameter of 2mm.

f. Permitted deviations from the above specifications are as follows :

Diameter of wire		Permissible Deviation
Over mm	Up to mm	mm
2.36	3.15	± 0.08
3.15	5.0	± 0.10

CONCRETE

10. Concrete posts and struts shall be reinforced concrete and shall comply with the following specifications:

a. Material

(1) Cement used in the manufacture shall be Portland ordinary or rapid hardening cement or Portland blast furnace cement. Where the owner specifies, sulphate - resisting cement or high alumina cement may be used.

(2) Aggregate shall not exceed 10mm nominal size maximum.

(3) Additives where used shall be in such proportion as to have no harmful effects on the setting, hardening or durability of the concrete.

(4) Reinforcement bars of all posts and struts shall consist of a prefabricated cage of four steel bars secured by mild steel stirrups of not less than 2.65mm diameter spaced at centres not exceeding 0.38 meters. The stirrups shall be close to the bars and parallel to the faces of the posts. The sizes of mild steel reinforcing bars shall be as follows:

(a) Reinforcement for intermediate posts shall be four mild steel bars of 8mm diameter.

(b) Reinforcement for straining posts shall be four mild steel bars of 10mm diameter.

(c) Reinforcement for struts shall be four mild steel bars of 8mm diameter.

(5) The cover of the concrete to the reinforcing bars shall be not less than 16mm. The reinforcing bars shall extend to within 25mm of each end of the unit. During manufacture the cage shall be controlled in position to ensure that this is achieved.

b. Moulds shall be so constructed that they remain rigid during the placing and compaction of the concrete to prevent the loss of water, grout or mortar from the concrete. They shall be such that the finished products are accurate within the limits specified and shall produce the required surface finish.

c. Finish shall be obtained by casting the unit in properly designed moulds. Where these are of wood they shall be closely-jointed wrought timber and the faces of the unit will be imprinted with the slight grain of the wrought timber and their joints. In addition, irrespective of the material used for the moulds, small surface blemishes caused by entrapped air or water may be expected, but the surface shall be free from honey combing or other large blemishes.

d. Proportioning: The cube strength of the concrete mix shall be not less than 25.5 N/mm^2 when tested after 28 days, or 17.2 N/mm^2 when tested after 7 days from manufacture. The above strengths may be expected from a mix composed of the part by weight of cement and not more than five parts by weight of graded aggregate, or equivalent by volume. The water/cement ratio should not exceed 0.55 by weight.

e. Mixing should be by mechanical mixer with the minimum quantity of clean water to provide for effective hydration, mixing and compaction, until there is a uniform distribution of the materials and the mass is uniform in colour and consistency.

f. Placing: The concrete shall be used as soon as possible after being mixed and no material shall be used in the work if it cannot be compacted fully without the addition of extra water. Vibration, tamping or other effective method shall be used to compact the concrete.

g. Maturing: Unless otherwise authorized by the owner in writing, no product shall be supplied until it has been allowed to mature under suitable conditions for a period of not less than 28 days.

h. Tests: The results of tests shall comply with the requirements outlined at sub-paragraph d. above.

i. The following permissible deviations shall apply to all the specified dimensions:

(1)	Cross Section	-	(-) 4mm to (+) 6mm
(2)	Length	-	(-) 12mm to (+) 38mm

The two long arises at the back of the posts shall be rounded, or sharp edges removed before dispatch. Tops shall be rounded. Except for cranked tops, posts shall be of the same cross section throughout their length, subject to the demoulding draw allowance of 4mm.

j. Intermediate Posts shall be 125mm x 100mm and shall have a vertical height of 3.2m overall. They shall have 10mm diameter holes for stirrup wires, and, if specified, 12mm diameter holes for fixing extension arms.

k. Straining Posts shall be 150mm x 150mm and shall have a vertical height of 3.2m overall. They shall be designed to provide a firm bearing for the struts which shall meet the straining posts at a point within the top third of the length of the straining post which is above ground. They shall have 12mm diameter holes for fixing eyebolts and, if specified, extension arms.

l. Cranked Tops to posts shall be at an angle of 40º to 45º to the vertical and shall be of such lengths as will increase the vertical height of the fence to 2.9m. They shall be cast with the posts. Cranked tops for intermediate posts shall taper to 100mm x 85mm at the top and shall be provided with 10mm diameter holes for stirrup wires for securing three lines of barbed wire 200mm apart. Cranked tops for straining posts shall taper to 150mm x 85mm at the top and shall be provided with 12mm diameter holes for straining eyebolts for securing three lines of barbed wire 200mm apart. Struts shall be 100mm x 100mm with parallel sides and 3.25m long.

m. Posts are to be set at not more than 3m centres.

n. Concrete surrounding the bases of the posts, struts and chain-link fencing shall be not leaner by weight than one part of cement to ten parts of graded aggregate of 40mm nominal maximum size, mixed with the minimum requisite quantity of clean water. The constituents shall be thoroughly mixed and the concrete placed in position and thoroughly compacted as soon as possible after mixing.

ERECTION – LINE AND LEVEL

11. The fence shall be so erected that on completion it is truly on the line of the boundaries set out by the owner. The top of the fence shall follow the profile of the ground. Any opening below or through the fence shall be protected in accordance with the owner's specifications. The strength of any opening so treated shall be not less than the strength of the remainder of the fence.

NB. The fence line must meet the specifications set out in paragraphs 7 & 8 above.

12. Straining posts shall be provided at all ends and corners of fences, at changes of direction or acute variations in level and at intervals not exceeding 66m on straight lines of fence. Intermediate posts shall be spaced at regular intervals not exceeding 3m. Holes for straining posts and gate post shall be not less than 0.45m x 0.45m on plan. Holes for intermediate posts shall be not less than 0.3m x 0.3m on plan, or if bored of equivalent area. All holes shall be 0.8m deep and shall have vertical sides. All posts shall be set in the ground to a depth of 0.76m except as varied to obtain the levels specified in Paragraph 11.

13. After insertion of posts, holes shall be filled not less than half their depth in concrete, well rammed as the filling proceeds and consolidated. After the concrete has hardened the remainder of the holes shall be backfilled, well rammed and consolidated. Holes for gate posts shall be filled completely to ground level with concrete, well rammed as the filling proceeds and consolidated.

14. Struts shall be fitted to all straining posts behind the chain-link fabric in the direction of the line of the fence. The bottom end shall be not less than 0.6m vertically below the ground line (graded). The holes in the ground to receive struts shall be not less than 0.6m x 0.3m on plant, with three vertical sides. After insertion of the struts, the holes shall be filled to not less than half their depth with concrete, well rammed as the filling proceeds and consolidated. The minimum cover on all faces and at the ends of the struts shall be 75mm. After the concrete has hardened the remainder of the holes shall be backfilled, well rammed and consolidated.

15. There shall be four evenly spaced rows of line wires. The top wire shall be doubled, making five line wires in all. The bottom line wire shall be close to the ground. Each line wire shall be strained tightly by means of an eyebolt strainer or winder at each straining post. Each line wire shall be secured to each intermediate post by a wire stirrup passed through a hole in the post and secured to the line wire by three complete turns on each side of the post.

16. The chain-link fencing shall be strained between each pair of straining posts by means of a stretcher bar. One of the top line wires shall be threaded through the appropriate adjacent rows of meshes, care being taken that no meshes in the rows are bypassed by the line wire except where deviation is necessary at the straining posts. The second top line wire shall be strained in front of the fencing. The fencing shall be attached to the top and bottom line wires by wire ties 450mm apart.

17. The bottom of the fencing shall be treated as follows:

    a. The bottom 0.2m of the fencing shall be buried vertically in the ground.

    b. The lower 0.15m shall be sunk into the concrete channel as defined at paragraph 7 above.

    c. The ground surface shall be reinstated.

18. Cranked tops or single extension arms shall point in the direction specified by the owner. Extension arms shall be supplied for concrete posts not having cranked tops

cast on. Where double extension arms are specified these shall consist of two single arms fixed, pointing in opposite directions, on opposite sides of each post. Where barbed wire is specified both inside and outside a fence having posts with cranked tops, single extension arms shall be used pointing in the opposite direction to that of the cast cranked top.

19. Three lines of barbed wire shall be provided. The lowest line shall be not more than 150mm above the top of the chain-link and the other equally spaced to a position not exceeding 50mm below the top of the extensions or cranked tops. The wire shall be attached by eyebolts to the cranked tops of the straining posts, or to the extension arms and shall be properly strained. On the intermediate posts they shall be secured to the cranked tops with stirrup wires. The barbed wire shall be fitted with one dropper at the centre of each bay, secured to the wires so that they cannot be bunched together.

20. All bolts to fencing shall be burred over.

21. Where gates are installed over other than hard, properly constructed and surfaced roadways, a concrete sill shall be formed carried up to within 50mm of the bottom of the gate. The top of the sill shall be at ground level.

22. Fittings for fencing shall be as follows:

    a. All fittings shall be galvanized.

    b. Cleats, one to each eyebolt, shall be of uniform size and shall consist of mild steel angle not less than 76mm x 50mm x 6.3mm.

    c. Eyebolt strainers shall consist of 10mm diameter bolt with welded eye sufficiently threaded and fitted with a nut and washer. Two-way eyebolt strainers shall have suitable ring nuts, fitted after the wires have been stained on one side.

    d. Stretcher bars shall consist of mild steel flat 25mm x 4.75mm. They shall be secured to the cleats by 8mm diameter steel bolts.

    e. Droppers for barbed wire shall be of mild steel not less than 25mm x 4.75mm thick with 38mm x 4.85mm half round staples for fastening the barbed wire to them.

Unquote

2. ADDITIONAL UNDERSTANDINGS

2.1 Although not stated in the verbatim section, it is understood that two roads are required for patrolling the security fence. A road is required both inside and outside the security fence and is to be level and a minimum, of 4.6m wide with a marl cap, i.e., no paving is necessary.

2.2 The security fence must be a minimum of 60m from the nearest vital equipment.

2.3 If the facility is built on an elevated pad, the security fence must be 30m from the toe of the berm. The slope from the toe of the berm to the security fence should not exceed 0.3m in 30m. The slope from the security fence to 30m past the security fence should not exceed 0.3m in 6.1m.

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.03

Revision No. 0

SAFETY & SECURITY – CATEGORY II FENCE

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1. SCOPE

This Standard refers to category II fence and is adopted verbatim from SAG SSD/2.

Quote

1. Fencing in use will conform to the following standard:

   a. The height of the fence including any cranked extension shall be not less than 2.9m above the ground line (graded).

   b. PVC coated chain-link fencing is to be used. The bond between the vinyl coating and the metal shall be not less than the cohesive strength of the vinyl. Extruded PVC shields or jacketed-type coatings are not acceptable. The colour of the PVC is to be dark green or black.

   c. The chain-link fence shall have 50mm mesh of 4mm diameter steel wire or heavier and twisted and barbed top and bottom selvages.

   d. Barbed wire shall consist of three rows of 2 strand 12.5 SWG galvanized 4 point thick-set barbed wire, secured to the posts with 14.5 SWG galvanized wire loops.

   e. The top lines of the barbed wire shall be fitted with droppers so that they cannot be bunched together.

   f. As far as possible the fence to run in straight lines. Where change of direction is necessary it should be not less than 130º.

   g. The fence to be sunk into a minimum 150mm concrete channel to prevent burrowing. The channel to be at a minimum 500mm below the ground line (graded).

   h. Where the fence is erected on a coast, the minimum distance from the fence to the high water mark shall be not less than 60m.

   i. A clear zone is to be created on either side of the fence to a minimum depth of 15m.

2. Where posts and struts are of metal they are to be as follows:

   a. Mild steel angle, or

   b. Steel tube, or

   c. Hollow Sections.

3. Specifications are as under:

a. Intermediate Posts

(1) Where mild steel angle is used the vertical length is to be 3.2m, the section to be 51mm x 51mm x 6.5mm.

(2) Where steel tube is used the vertical length is to be 3.2m, the section to be 48mm outside diameter x 3.2mm.

(3) Where hollow section is used the length is to be 3.2m, the section to be 38mm x 38mm x 4mm.

b. Straining Posts

(1) Where mild steel angle is used the vertical length is to be 3.2m, the section to be 63.5mm x 63.5mm x 8mm.

(2) Where steel tube is used the vertical length is to be 3.2m, the section to be 60mm outside diameter x 4mm.

(3) Where rectangular hollow section is used the vertical length to be 3.2m, the section to be 51mm x 51mm x 4mm.

c. Struts

(1) Where mild steel angle is used the vertical length is to be 3.2m, the section to be 44.5mm x 44.5mm x 4.67mm.

(2) Where steel tube is used the vertical length is to be 3.2m, the section to be 48mm outside diameter x 3.2mm.

(3) Where rectangular hollow section is used the vertical length to be 3.2m, the section to be 38mm x 38mm x 3.2mm.

4. All struts should have a tie bar not less than 0.75m long securely fixed to strut and straining post with bolts not less than 7.9mm diameter. The tie bar should have the same section as the strut.

5. All metal used shall be galvanized after manufacture.

6. The bottom of all angle steel posts and struts shall be spragged and tubes and hollow sections shall have base plates 150mm x 150mm x 3mm to provide a key in concrete bases. The ends of all hollow steel posts shall be capped and sealed.

7. Steel extension arms for steel posts shall be mild steel angle 44.5mm x 44.5mm x 4.67mm and galvanized. Each arm shall be holed to take half round staples for fixing three lines of barbed wire 200mm apart and shall be provided with two 9.5mm diameter galvanized bolts.

8. Steel extension arms for tubular, circular and rectangular hollow posts shall be welded on or shall be a continuation of the tubular post. The extension shall form an angle of 40º or 45º with the vertical line of the post and shall be such a length as will increase the vertical height of the fence to 2.9m.

9. Where concrete posts are specified the following standards will be observed:

a. Materials

(1) Cement used in the manufacture shall be Portland ordinary or rapid hardening cement or Portland blast furnace cement. Where the owner specifies, sulphate resisting or high alumina cement may be used.

(2) Aggregate shall not exceed 10mm nominal size maximum.

(3) Additives where used shall be in such proportion as to have no harmful effects on the setting, hardening or durability of the concrete.

(4) Reinforcement bars of all posts and struts shall consist of a prefabricated cage of four steel bars secured by mild steel stirrups of not less than 2.65mm diameter spaced at centres not exceeding 38mm. The stirrups shall be close to the bars and parallel to the faces of the posts. The sizes of mild steel reinforcing bars shall be as follows:

(a) Reinforcement for intermediate posts shall be four mild steel bars of 8mm diameter.

(b) Reinforcement for straining posts shall be four mild steel bars of 10mm diameter.

(c) Reinforcement for struts shall be four mild steel bars of 8mm diameter.

(5) The cover of the concrete to the reinforcing bars shall be not less than 16mm. The reinforcing bars shall extend to within 25mm of each end of the unit. During manufacture the cage shall be controlled in position to ensure that this is achieved.

b. Moulds shall be so constructed that they remain rigid during the placing and compaction of the concrete to prevent the loss of water, grout or mortar from the concrete. They shall be such that the finished products are accurate within the limits specified and shall produce the required surface finish.

c. Finish shall be obtained by casting the unit in properly designed moulds. Where these are of wood they shall be close-jointed wrought timber and the faces of the unit shall be imprinted with the slight grain of the wrought

timber and their joints. In addition, irrespective of the material used for the moulds, small surface blemishes caused by the entrapped air or water may be expected, but the surfaces shall be free from honey combing or other large blemishes.

d. Proportioning. The cube strength of the concrete mix shall be not less than 25.5 N/mm^2 when tested after 28 days or 17.2 N/mm^2 when tested after 7 days. The above strengths may be expected from a mix composed of one part by weight of cement and not more than five parts by weight of graded aggregate or equivalent by volume. The water/cement ratio should not exceed 0.55 by weight.

e. Mixing should be by mechanical mixer with the minimum quantity of clean water to provide for effective hydration, mixing and compaction, until there is a uniform distribution of the materials and the mess is uniform in colour and consistency.

f. Placing. The concrete shall be used as soon as possible after being mixed and no material shall be used in the work if it cannot be compacted fully without the addition of extra water. Vibration, tamping or other effective method shall be used to compact the concrete.

g. Maturing. Unless otherwise approved by the owner in writing, no product shall be supplied until it has been allowed to mature under suitable conditions for a period of not less than 28 days.

h. Tests. The results of tests shall comply with the requirements outlined at sub-paragraph d. above.

i. The following permissible deviations shall apply to all the specified dimensions:

(1)	Cross Section	-	(-) 4mm to (+) 6mm
(2)	Length	-	(-) 12mm to (+) 38mm

The two long arises at the back of the posts shall be rounded, or sharp edges removed before dispatch. Tops shall be rounded. Except for cranked tops, posts shall be of the same cross section throughout their length, subject to the demoulding draw allowance of 4mm.

j. Intermediate Posts shall be 125mm x 100mm and shall have a vertical height of 3.2m overall. They shall have 10mm diameter holes for stirrup wires and, if specified, 12mm diameter holes for fixing extension arms.

k. Straining Posts shall be 150mm x 150mm and shall have a vertical height of 3.2m overall. They shall be designed to provide a firm bearing for the struts which shall meet the straining posts at a point within the top third of the

length of the straining post which is above ground. They shall have 12mm diameter holes for fixing eyebolts and, if specified, extension arms.

l. Cranked Tops to posts shall be at an angle of 40º to 45º to the vertical and shall be of such lengths as will increase the vertical height of the fence to 2.9m. They shall be cast with the posts. Cranked tops for intermediate posts shall taper to 100mm x 85mm at the top and shall be provided with 10mm diameter holes for stirrup wires for securing three lines of barbed wire 200mm apart. Cranked tops for straining posts shall taper to 150mm x 85mm at the top and shall be provided with 12mm diameter holes for straining eyebolts for securing three lines of barbed wire 200mm apart. Struts shall be 100mm x 100mm with parallel sides and 3.25m long.

m. Posts to be set at not more than 3m centres.

n. Concrete surrounding the bases of the posts, struts and chain-link fencing shall be not leaner by weight than one part of cement to ten parts of graded aggregate of 40mm nominal maximum size, mixed with the minimum requisite quantity of clean water. The constituents shall be thoroughly mixed and the concrete placed in position and thoroughly compacted as soon as possible after mixing.

ERECTION – LINE AND LEVEL

10. The fence shall be so erected that on completion it is truly set out on the line of the boundaries set out by the owner. The top of the fence shall follow the profile of the ground. Any opening below or through the fence shall be protected in accordance with the owner's specifications. The strength of any opening so treated shall be not less than the strength of the remainder of the fence.

NB. The fence line must meet the specifications set out in paragraphs 1h and 1i above.

11. Straining posts shall be provided at all ends and corners of fences, at changes of direction or acute variations in level and at intervals not exceeding 66m on straight lines of fence. Intermediate posts shall be spaced at regular intervals not exceeding 3m. Holes for straining posts and gate post shall be not less than 0.45m x 0.45m on plan. Holes for intermediate posts shall be not less than 0.3m x 0.3m on plan, or if bored, of equivalent area. All holes shall be 0.8m deep and shall have vertical sides. All posts shall be set in the ground to a depth of 0.76m except as varied to obtain the levels specified in Paragraph 10 above.

12. After insertion of posts, holes shall be filled not less than half their depth in concrete, well rammed as the filling proceeds and consolidated. After the concrete has hardened the remainder of the holes shall be backfilled, well rammed and consolidated. Holes for gate posts shall be filled completely to ground level with concrete, well rammed as the filling proceeds and consolidated.

13. Struts shall be fitted to all straining posts behind the chain-link fabric in the direction of the line of the fence. The bottom end shall be not less than 0.6m vertically below the ground line (graded). The holes in the ground to receive struts shall be not less than 0.6m x 0.3m on plan, with three vertical sides. After insertion of the struts, the holes shall be filled to not less than half their depth with concrete, well rammed as the filling proceeds and consolidated. The minimum cover on all faces and at the ends of the struts shall be 75mm. After the concrete has hardened the remainder of the holes shall be backfilled, well rammed and consolidated.

14. There shall be four evenly spaced rows of line wires. The top wire shall be doubled, making five line wires in all. The bottom line wire shall be close to the ground. Each line wire shall be strained tightly by means of an eyebolt strainer or winder at each straining post. Each line wire shall be secured to each intermediate post by a wire stirrup passed through a hole in the post and secured to the line wire by three complete turns on each side of the post.

15. The chain-link fencing shall be strained between each pair of straining posts by means of a stretcher bar. One of the top line wires shall be threaded through the appropriate adjacent rows of meshes, care being taken that no meshes in the rows are bypassed by the line wire except where deviation is necessary at the straining posts. The second top line wire shall be strained in front of the fencing. The fencing shall be attached to the top and bottom line wires by wire ties 450mm apart.

16. The bottom of the fencing shall be treated as follows:

    a. The bottom 0.2m of the fencing shall be buried vertically in the ground.

    b. The lower 0.15m shall be sunk into the concrete channel as defined at paragraph 1g above.

    c. The ground surface shall be reinstated.

17. Cranked tops or single extension arms shall point in the direction specified by the owner. Extension arms shall be supplied for concrete posts not having cranked tops cast on. Where double extension arms are specified these shall consist of two single arms fixed, pointing in opposite directions, on opposite sides of each post. Where barbed wire is specified both inside and outside a fence having posts with cranked tops, single extension arms shall be used pointing in the opposite direction to that of the cast cranked top.

18. Three lines of barbed wire shall be provided. The lowest line shall be not more than 150mm above the top of the chain-link and the other equally spaced to a position not exceeding 50mm below the top of the extensions or cranked tops. The wire shall be attached by eyebolts to the cranked tops of the straining posts, or to the extension arms and shall be properly strained. On the intermediate posts they shall be secured to the cranked tops with stirrup wires. The barbed wire shall be fitted with one dropper at the centre of each bay, secured to the wires so that they cannot be bunched together.

19. All bolts to fencing shall be burred over.

20. Where gates are installed over other than hard, properly constructed and surfaced roadways, a concrete sill shall be formed carried up to within 50mm of the bottom of the gate. The top of the sill shall be at ground level.

21. Fittings for fencing shall be as follows:

    a. All fittings shall be galvanized.

    b. Cleats, one to each eyebolt, shall be of uniform size and shall consist of mild steel angle not less than 76mm x 50mm x 6.3mm.

    c. Eyebolt strainers shall consist of 10mm diameter bolt with welded eye sufficiently threaded and fitted with a nut and washer. Two-way eyebolt strainers shall have suitable ring nuts, fitted after the wires have been stained on one side.

    d. Stretcher bars shall consist of mild steel flat 25mm x 4.75mm. They shall be secured to the cleats by 8mm diameter steel bolts.

    e. Droppers for barbed wire shall be of mild steel not less than 25mm x 4.75mm thick with 38mm x 4.85mm half round staples for fastening the barbed wire to them.

<u>Unquote</u>

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.04

Revision No. 0

SAFETY & SECURITY – CATEGORY III FENCE

PROPRIETARY INFORMATION

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1. SCOPE

This Standard refers to category III fence and is adopted verbatim from SAG SSD/3.

Quote

1. Fencing in use shall conform to the following specifications:

a. The height of the fence shall not be less than 2m.

b. Galvanized chain-link fencing is to be used.

c. The chain-link fabric shall have 50mm mesh of 4mm diameter steel wire or heavier and twisted and barbed top and bottom selvages.

d. As far as possible the fence to run in straight lines. Where a change of direction is necessary, it should be not less than 130º.

e. Line posts shall be steel standard pipe of not less than 60mm outside diameter and weighing not less than 5.45 kg/m.

f. End, corner and pull posts shall be steel standard pipe of not less than 75mm outside diameter weighing not less than 8.6 kg/m.

g. Hinged gate posts shall be steel standard pipe with the following diameters and weights for the nominal widths of a single gate or of one leaf of a double gate:

Nominal Gate Width	Minimum outside Diameter	Minimum Weight
1.8m and less	75mm	8.6 kg/m
Over 1.8m and up to 4m	100mm	13.6 kg/m

h. Post tops and other fittings and hardware shall match the material chosen for the posts.

i. Top and bottom tension wires between posts shall be at least 5mm diameter steel. Tie wire shall be 4mm diameter steel.

j. Stretcher bars shall be steel, not less than 5mm x 19mm or equivalent cross-sectional area and of lengths equal to the full height of the fabric to be used.

k. Braces shall be steel pipe of at least 40mm outside diameter and weighing not less than 2 kg/m.

l. Bolts, nuts and other hardware shall be galvanized steel.

m. Bolts are to be burred over to prevent dismantling.

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REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.05

Revision No. 0

SAFETY & SECURITY – CATEGORY III FENCE ALTERNATIVE

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1. SCOPE

This Standard refers to category III fence – alternative and is adopted verbatim from SAG SSD/4.

Quote

1. The following minimum specifications shall be observed.

   a. Fence to be constructed of Class I Zinc-coated woven wire fabric.

   b. Minimum height to be 1.5m above ground line (graded).

   c. Fabric shall have 9 horizontal wires at 120mm centres with vertical stays 150mm apart.

   d. Top and bottom wires shall be 3.55mm diameter and the intermediate wires and vertical stays 2.5mm diameter.

   e. Where barbed wire is fitted, it shall consist of zinc-coated two strand wire with 2.5mm diameter 4 point 1.9mm diameter round barbs spaced not more than 130mm apart.

   f. Tie and brace wire shall be 4.11mm diameter zinc-coated steel.

   g. Tie wire for tying bottom strand of barbed wire to top of woven wire fence fabric shall be 2.7mm diameter zinc-coated carbon steel wire.

   h. Terminal, corner and brace posts and diagonal braces shall be standard pipe of not less than 73mm outside diameter and shall weigh not less than 8.62 kg/m.

   i. Hinged gate posts shall be standard pipe of not less than 100mm diameter and shall weigh not less than 10.79 kg/m.

   j. Post tops and other fittings and hardware shall be galvanized steel.

   k. Stretcher bars shall be structural steel, galvanized, and be not less than 6.5mm x 19mm or equivalent cross-sectional area and of lengths equal to the full height of the fabric to be used. The bars shall be attached to the fabric by clamps or other approved mechanical means. One stretcher bar shall be provided for each gate, gate hinge and terminal post and two for each corner post and pull post assembly. Fabric and stretcher bars to be attached to posts at intervals not exceeding 400mm.

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REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.06

Revision No. 0

SAFETY & SECURITY – GATES IN PERIMETER FENCES

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1. SCOPE

This Standard refers to gates in standard security perimeter fences and is adopted verbatim from SAG SSD/5.

Quote

1. Gates are to be reduced to a minimum compatible with the requirements for control of access and safety and emergency procedures. They are to be designed to eliminate as far as possible any increased facilities for unauthorized entry.

2. The following minimum specifications are to apply:

   a. Single gates shall be used to fill opening up to 2m in width.

   b. Double leaves of equal width shall be used to fill openings up to 8m in width.

   c. The tops of the gates shall be leveled with the tops of the fencing.

   d. The tops of the barbed wire shall be leveled with the top of the barbed wire fencing.

   e. The bottom of the gates shall have a ground clearance of not more than 50mm at the meeting point unless the site conditions make a greater clearance necessary.

   f. When closed, the space between the meeting stiles shall not be more than 12mm.

   g. For gates up to 2m in width the following specifications are to apply.

      (1) Galvanized steel tubes 48.3mm x 3.2mm thickness hollow section are to be used for the frames.

      (2) Galvanized steel tubes 42.2mm x 3.2mm thickness circular hollow section are to be used for the braces.

   h. For gates over 2m in width the following specifications are to apply:

      (1) Galvanized steel tubes 60.3mm x 3.2mm thickness hollow section are to be used for the frames.

      (2) Galvanized steel tubes 48.3mm x 3.2mm thickness circular hollow section are to be used for the braces.

3. Construction of gates shall be as follows:

   a. Gates shall be of all-welded construction.

b. Gates and fittings shall be galvanized after manufacture.

c. The frame shall be square with all corners mitred or saddled. Bent tubes or radiused corners shall not be used.

d. Vertical braces shall be fixed at not more than 1.2m centres and there shall be one central horizontal brace.

e. Infilling of gates shall consist of 4mm galvanized mild steel wire fabric of not larger than 50mm x 50mm square mesh welded at all intersections with the wire running horizontally and vertically.

f. Infilling shall be fixed to the frame by galvanized clamp bars 25mm x 4.75mm stitch-welded to the frame up to 25mm from each junction and the wire fabric, with a selvedge all round, clamped to this with a 25mm x 4.75mm clamp bar bolted at not more than 0.30m centres with 25mm x 8mm sherardized bolts and nuts.

g. The outside of the gate shall be flushed.

h. Chain-link infilling of a similar mesh and gauge to the fencing may be used if specified by the owner.

i. Extensions arms and barbed wire. The hanging and shutting stiles shall be extended vertically upwards 0.45m to carry three lines of barbed wire.

j. Each vertical brace shall have a 0.45m extension arm, not cranked, above the frame, welded and saddled over it and drilled to take and include half round fasteners for securing the barbed wire.

k. The barbed wire shall comply with the specifications laid down in SSD/l and shall be stretched hand tight and secured to lugs at the stiles.

l. Hinges: The bottom hinge shall be of the heel and socket type. For gates up to 2m wide the plate shall be 12.7mm thick and the pin shall have a minimum diameter of 19mm. For gates over 2m wide and up to 4m wide the plate shall be 25mm thick and the pin shall have a minimum diameter of 25mm.

m. The top hinge for gates up to 2m wide shall be made from 12.7mm thick plate with a 12.7m diameter pin and for gates over 2m wide and up to 4m wide shall be made from 25mm thick plate with 25mm diameter pin.

n. The female half of the hinge pin shall be welded to the gate frame and the plate carrying the pin shall be bolted to the gate posts and inverted.

o. Drop bolts and slam plates: Mild steel drop bolts 20mm in diameter shall be so fitted to all gate frames (and therefore to each leaf of a double gate) that they cannot be removed. Corresponding sleeves to receive the bolts shall be set securely in the ground and concreted in.

p. The last closing leaf of a pair of double gates shall incorporate a stop which prevents one drop bolt being lifted when the gates are closed.

q. Double gates shall be provided with a slam plate of not less than 12mm thickness.

r. Locking devices: Each gate shall be fitted with a locking bar not less than 38mm x 9.5mm flat secured in a frame 38mm x 9.5mm flat welded to the gate frame at the junction of the horizontal brace on the shutting stile.

s. For single gates the locking bar shall shoot into a socket on the gate post.

t. For pairs of gates the locking bar shall shoot into a staple not less than 25mm x 4.7mm and all locking bars shall be fitted with 63.5mm 7-level galvanized padlocks.

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REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.07

Revision No. 0

SAFETY & SECURITY – GATES IN WOVEN WIRE FABRIC FENCES

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1. SCOPE

This Standard refers to gates in woven fabric fences and is adopted verbatim from SAG SSD/5A.

Quote

1. Where woven wire fabric fences are constructed the following specifications for gates will apply:

   a. Where pedestrian access only is required the width of the gate shall not exceed 1.5m.

   b. Where vehicular access is required the width of the gate shall not exceed 7m, the gate to consist of equal width double leaves.

   c. Gate frames shall be galvanized standard pipe of not less than 48mm outside diameter weighing at least 4.05 kg/m.

   d. Frames shall be cross-braced with 20mm diameter adjustable truss rods, and the corners shall be welded.

   e. The wire fabric shall be attached to the gate frames by stretcher bars and tie wires, as for the fence construction and tension connections shall be spaced 300mm apart.

   f. Gates to be hung on minimum two galvanized steel hinges not less than 75mm wide, clamped securely to the gate post and allowing the gate to swing back against the fence.

   g. Hinges shall be designed and so arranged that the gate cannot be removed without mechanically dismantling the hinges.

   h. Where barbed wire is fitted to the fence, extension arms are to be installed on the end posts to carry strands of barbed wire horizontally aligned with those of the adjacent fence, or the vertical frame members may be extended to carry the barbed wire.

   i. Gates shall be secured with padlocks and to this and each gate shall be fitted with a latch with padlock eyes of the forked or plunger type, operated from either side of the gate. Where double gates are installed, the gates shall be fitted with stops and a centre rest and catch.

Unquote

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.08

Revision No. 0

SAFETY & SECURITY – BARBED "S" WIRE CONCERTINA

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1. SCOPE

This Standard refers to Barbed S Wire Concertina and is adopted verbatim from SAG SSD/6.

Quote

1. This Directive lays down the minimum specifications of Barbed S Wire Concertina for use in installations within the Kingdom. Barbed S Wire will not be installed in any industrial installation without the express, written authority of the Secretariat, (HCIS).

2. The Barbed S Wire Concertina shall consist of single-strand preformed spring steel line wire fitted with Barbed S wire. The coil shall be not less than 0.70m or more than 0.75m in diameter and shall consist of 52 spiral turns connected by steel clips so as to form a cylindrical diamond-pattern when extended to a coil length of a maximum 15m.

3. One end turn of the coil shall be fitted with four bundling wires for securing the coil when compressed and each end turn shall be fitted with two steel carrying loops. The Concertina shall maintain circular shape when extended to the required length and shall not deform permanently when weight is applied to the stretched wire and, when released shall retract into a compact coil.

4. The line wire shall be made from spring steel with a minimum diameter of 2.5mm.

5. Steel strip for Barbed S Wire Concertina shall be made from low carbon steel that has been suitably tempered. It shall have a thickness of 0.50mm and shall be electro-galvanized with zinc to a thickness of not less than 0.0025mm on each side. Hot dip galvanizing may be used providing the specified thickness is maintained.

6. Carrying loops shall be of galvanized steel of not less than 4.8mm diameter.

7. Clips for Barbed S Wire Concertina shall be made from steel not less than 1.7mm thick.

8. The line wire together with clinched-on Barbed S Wire shall be preformed to fall naturally into coils and shall not show a wavy condition nor form a figure eight. Spliced joints shall be neatly and securely made and shall have a breaking strength of not less than 5780N. There shall be no more than one spliced line wire joint to a single Concertina.

9. The two outer turns at each end of the concertina shall be clipped together at four points equally spaced. The end turn shall be connected to its adjacent turn with two clips spaced at not more than 150mm centres. The line wire shall be connected by clips every 120° ± 10° on the circumference. Each turn shall be connected to the succeeding turn by the remaining clip.

10. The steel clips when clinched on the line wire shall allow the coil to form the diamond pattern mesh without permanent distortion of the line wire at the clips. The clips shall be closed sufficiently to prevent the line wire from pulling free.

11. Wire used in the manufacture of the concertina shall be free of links and bends. Shearing and bending shall be neat and accurate and barbs shall be correctly formed and sharp.

12. The mounting of Barbed S Wire Concertina on Standard Security Perimeter Fencing shall be as shown in Annex C to this SSD. Other configurations are not considered effective and are not to be used.

Unquote

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.09

Revision No. 0

SAFETY & SECURITY – SECURITY & EMERGENCY EXIT DOORS

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1. SCOPE

This Standard refers to security and emergency exit doors and is adopted verbatim from SAG SSD/7.

Quote

1. Doors designated as security doors or emergency exit doors in buildings housing sensitive or vital equipment are to be constructed so as to resist forcible attack. The following minimum specifications are to apply.

   a. Door to be minimum 10mm mild steel.

   b. Hung on three heavy duty hinges.

   c. Frames to be not less than 115mm x 75mm x 10mm mild steel angles.

   d. Frames to be recessed from the external face of the wall and secured to the jambs with four 230mm flanged building-in lugs welded to each side of the frame. The frame to be properly grouted after fixing.

   e. The hinges to be continuously welded to the door frame and the door and their construction to be such as to prevent removal of the hinge pins.

   f. Whenever possible, the doors to be outward opening.

   g. On an outward opening single leaf door, any clearance between the locking stile and the door frame to be covered by an external mild steel plates 150mm x 230mm x 10mm positioned to cover the lock bolt in the thrown position.

   h. On double-leaf unrebated doors, external and internal covering strips extending the full length of the meeting stiles are to be fitted.

   i. All locks are to be suitable security locks as specified in SSD/8.

2. Where operational conditions preclude the use of mild steel doors, a solid core wooden door not less than 50mm thick may be installed.

3. All security doors are to be fitted with a stout hasp and staple of the REVOBAR type.

4. All security and emergency doors are to comply with the specifications laid down in the latest edition of NFPA 80 and NFPA 101or better.

Unquote

2. ADDITIONAL UNDERSTANDINGS

2.1 It is understood that the requirements of this specification apply to the doors of electrical switchgear buildings, communication buildings, and control buildings of vital facilities, and to buildings holding any vital or sensitive equipment.

2.2 The REVOBAR type of hasp and staple protects the attaching screws from tampering when the hasp is closed over the staple.

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.10

Revision No. 0

SAFETY & SECURITY – LOCKS USED ON SECURITY DOORS, PERIMETER FENCES AND EMERGENCY EXITS

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1. SCOPE

This Standard refers to locks used on the security and emergency exit doors and is adopted verbatim from SAG SSD/8.

Quote

1. Locks used for perimeter fences and control of access to areas holding sensitive or vital equipment are to have the following characteristics:

a. Doors

(1) Be resistant to forcing

(2) Have five effective levers.

(3) Have minimum bolt throw of 16mm.

(4) Be resistant to drilling and the bolt to sawing.

(5) Fixing screws to be hidden when the door is closed.

(6) Box striking plate to be used.

(7) Lock to be morticed into fabric of door.

(8) Have deadlock action.

(9) Master key systems not to be used.

(10) Pin tumbler, disc tumbler and warded locks are not to be used.

(11) For double doors, sliding doors and folding doors, locks to have hook-bolt deadlock action.

(12) Door and frame to be fitted with stout hasp and staple of the REVOBAR type.

b. Emergency Exits

(1) Where panic bolts are fitted they are to:

(a) Afford resistance against forcible attack.

(b) Bolt to be spring-loaded in the opening direction.

(c) Be incorporated in any alarm detection system installed.

(2) Where a high quality security deadlock is installed, the door to be fitted with an electrically secured, quick release magnetic locking device. This device must meet the requirements of latest edition of NFPA 80 and NFPA 101 or better.

(3) Where the emergency exit is located in a Security area, the door is to be fitted with an automatic release system of the REDLAM Alarm Bolt type or better.

(4) Whatever the device installed, it must be incorporated into any alarm detection system.

c. Perimeter Fence fitted with security padlock

(1) Deadlock action.

(2) Minimum five effective levers

(3) Shackle to be resistant to sawing.

(4) Shackle to be resistant to being pulled away from body.

(5) Body to be resistant to sawing, drilling, chiseling or hammering.

(6) Body to protect shackle on three sides.

(7) Key to be withdrawn only when the lock is in the "locked" position.

(8) Shackle to be minimum 10mm diameter.

(9) Shackle to have maximum clearance of 16mm.

Unquote

2. ADDITIONAL UNDERSTANDING

2.1 The locks of this specification are to be used in conjunction with the doors of TES-S-101.09

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No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
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TES-S-101.11

Revision No. 0

SAFETY & SECURITY – BUILDINGS HOUSING VITAL EQUIPMENT

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1. SCOPE

This Standard refers to requirements for buildings housing vital equipment and is adopted verbatim from SAG SSD/9.

Quote

1. Where buildings are designated as housing sensitive or vital equipment the following minimum standards of construction are to be met.

   a. Walls are to be of solid structural material e.g. concrete block or concrete, not less than 250mm thick including where necessary a 50mm recess.

   b. Floors are to be constructed of reinforced concrete not less than is 150mm thick.

   c. Ceilings are to be constructed of reinforced concrete not less than 150mm thick.

   d. Windows are not to be built-in to the building.

   e. Ventilation in buildings housing sensitive or vital equipment shall be by air conditioning. The air conditioning units will be located on the outside of the building and be protected by caging or similar devices as specified in SSD/10.

   f. Doors will be kept to a minimum compatible with the requirements of safety and the smooth functioning and efficient operation of the equipment. Where doors are included, they are to meet the requirements for security and emergency doors as laid down in SSD/7.

2. Where sensitive or vital equipment are housed in large multi-purpose buildings, e.g. switchgear, generators, pumps etc. in warehouses, they should be bricked in or caged in heavy gauge mesh to secure them from the remainder of the building.

Unquote

2. ADDITIONAL UNDERSTANDING

2.1 It is understood that the requirements of this specification apply to buildings housing electrical switchgear, communication equipment, and diesel generators of vital facilities, and to all buildings containing facilities vital to the function of the plant.

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.12

Revision No. 0

SAFETY & SECURITY – PROTECTION OF VITAL EQUIPMENT

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1. SCOPE

This Standard refers to requirements for the protection of vital equipment and is adopted verbatim from SAG SSD/10.

Quote

1. Where the vital equipment such as transformers are in exposed position, they are to be protected by caging as follows:

   a. Heavy gauge mesh, e.g. 12 - 8 gauge, is to be used.

   b. The narrow dimension of the mesh not to exceed 13 mm.

   c. The mesh to be welded to a frame of mild steel angles spot welded at 250mm centres.

   d. Doors to gates are to be fitted so that the clearance between the gate and the frame and between stiles of a double gate does not exceed 13mm.

   e. The roof to slope at a minimum angle of 30º sloping away from the building, to allow any obstacle or missile to roll off.

   f. The cage to be incorporated in any alarm system in use.

   g. The doors to be secured by suitable security locks.

   h. Where gates are installed, they are to be secured by an approved security padlock and bar.

2. Where caging is not appropriate, equipment are to be protected by the use of a close-boarded wooden fence built at least 500mm higher than the equipment it is protecting. The roof is to be constructed as for caging.

3. Doors giving access to vital equipment are to be opened outwards unless specified otherwise, by the owner, for purposes of safety.

4. Where the structure housing vital equipment forms a part of a compound, the 30º angle roof should slope away from the other building.

5. Doors are to be fitted with a stout hasp and staple of the REVOBAR type.

6. The door size should be determined by the size of the largest piece of equipment housed.

Unquote

2. ADDITIONAL UNDERSTANDINGS

2.1 The requirements of this specification apply to vital equipment in exposed positions on site which also need protection, although not necessarily by the use of heavy gauge mesh. HCIS will review proposals for protection of exposed equipment on site in relation to their locations, but in general the SSD/TES-S-101 applies to all exposed vital equipment unless alternative means of protection have been approved by HCIS.

2.2 All vital equipment outside the security perimeter fence must be protected. The additional criteria for the determination of the protection to be afforded for the equipment that is to be protected outside the security perimeter fence is a function of equipment isolation and distance from the fence.

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.13

Revision No. 0

SAFETY & SECURITY – POWER SUPPLIES

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1. SCOPE

This Standard refers to requirements for power supplies and is adopted verbatim from SAG SSD/11.

Quote

1. Wherever possible, installation should have their own power generating plant. Where such plant is installed, it is to be sufficient to meet:

   a. All requirements for plant operations, communications and domestic and security lighting.

   b. Stand-by capacity for demands as outlined above.

   c. Emergency generating capacity sufficient to maintain communications, safety and security lighting in addition to any power required for controlled shut-down procedures.

2. Generators should be housed in solidly constructed buildings as specified in SSD/9.

3. Transformers are to be protected in accordance with the specifications as laid down in SSD/10, and are to be located adjacent to the power generating units at ground level.

4. Fuel for power generating equipment is to be stored and protected in accordance with current safety standards. It is to be located adjacent to the equipment it serves.

5. All cable ducts and all fuel lines are to be laid underground and come up inside buildings.

MAINS INPUT

6. Where mains input is used in the installation, there is to be a stand-by generating system sufficient to:

   a. Meet all the requirements for plant operations, communications, domestic and security lighting in the event of a mains power failure.

   b. Have emergency generating capacity sufficient to maintain communications and safety and security lighting in addition to that required to meet the demands for controlled shut-down procedures.

7. The mains input, switches and associated equipment are to be housed in solidly constructed buildings as specified in SSD/9. The buildings are to be located not less than 60m from the perimeter fence. Under no circumstances will a building power generating equipment or power distributing equipment form part of a perimeter fence.

8. Stand-by generators should be housed in solidly constructed buildings as per SSD/9 and be located adjacent to the mains input building utilizing the same switchgear.

9. Transformers are to be housed in solidly constructed buildings as per SSD/10 and be located adjacent to the stand-by power generating units at ground level.

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REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.14

Revision No. 0

SAFETY & SECURITY – COMMUNICATIONS

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1. SCOPE

This Standard refers to requirements for communications and is adopted verbatim from SAG SSD/12.

Quote

1. First line communications are to be by land line, using a common-user facility. Internal junction chambers for the land line are to be kept to a minimum, compatible with efficient and effective operations of the equipment.

2. Where it is essential, those additional communication facilities are installed then it should be a VHF radio system. The radio system is to have a minimum four channels allocated as follows:

    a. Operations
    b. Fire Crews
    c. Security
    d. Emergency Services

3. Sufficient hand-held radios are to be available to meet the demands of all users plus a back-up stock of spares for use in the event of an emergency, requiring the assistance of outside agencies, e.g. Fire Services, Security Services.

4. The communications systems are to have their own stand-by generators with sufficient capacity to meet the demands of a fully operational land line system and radio system in the event of a total power failure.

5. In addition to stand-by power, the communications systems are to have battery-generated emergency power sufficient to maintain the radio system fully as well as reduced land line system for a minimum period of 12 hours.

6. Where there is an Emergency Room in existence, the main console should be housed in a room adjacent to the Emergency Operations Room.

7. Telecommunications Department personnel should be requested to reduce the number of junction chambers to the absolute minimum. These chambers should be:

    a. Buried underground
    b. Chamber concrete lined
    c. Secured by heavy duty steel covers

8. Where land line communications are considered vital, a separate specification will be issued by the Secretariat for the protection of the system.

9. It is essential that any communications system installed is properly and thoroughly maintained. This should be part of the original bid specification.

Unquote

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
3					

TES-S-101.15

Revision No. 0

SAFETY & SECURITY – SECURITY LIGHTING SYSTEMS

PROPRIETARY INFORMATION

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1. SCOPE

This Standard refers to requirements for security lighting system and is adopted verbatim from SAG SSD/13.

Quote

1. Perimeter Lighting: Where security lighting is installed at perimeter security fences, the following minimum standards are required:

   a. Lights are to be mounted. 3 meters above the ground level at the fence line.

   b. Distance between lights is not to exceed 10 meters,

   c. Lights are to be placed 10 meters inside the fence line.

   d. Lights are to face outwards, illuminating the fence, the outside area, and the inside area to 3 meters from the fence line. The lower edge of the beam may be adjusted by mounting a shield on the underside of the luminary.

   e. The illuminance is to exceed that of the surrounding area, including street lighting.

   f. High pressure sodium discharge lamps should be used having the following characteristics:

      (1) The lamp is to be tubular horizontal burning.

      (2) Ballast and control gear are to be compatible with power supply at place of installation.

      (3) The lamp is to be 400w with a lighting design lumen output of 45000 lumens.

      (4) The restrike time on interruption of the power supply is to be no more than one minute.

      (5) The lamp is to take no more than five minutes to reach full brightness.

      (6) Lumen output beyond 2000 hours operation is not to decrease by over 2% per 1000 hours operation.

   g. Lamps are to be mounted in luminaries with the following specifications:

      (1) The mounting is to be adjustable in both the vertical and the horizontal planes. Stirrup mounts of galvanized steel or knuckled collar swivel mounts of galvanized steel or cast aluminium are to be used.

(2) The reflector is to give an asymmetric beam.

(3) The body is to be of heavy duty construction of a material which is corrosion resistant to elements present at the place of installation.

(4) The body is to fully protect the optical and electrical systems from outside elements.

(5) Front glass is to be heat and impact resistant. The glass is to be both hinged and completely removable. It should be secured with quick release key operated screws.

(6) It is to be suitably earthed.

h. In coastal areas, or areas subject to fog, mist or low cloud, low pressure sodium discharge lamps are to be used with a minimum lighting design output of 20000 lumens.

i. The control gear and ballast are to be easily to accessible to maintenance personnel. A remote control gear box or integral ballast tray may be used, and both must be protected and secured against unauthorized access. Where control gear is remote from the luminary, lamp igniter distances are not to exceed the manufacturer's recommended maximum.

j. The ballast must reliably start the lamp in ambient temperatures down to -5°C.

k. The system is to have robust fittings and armoured cables. Cables are to be buried underground, rising inside masts.

l. Each lamp is to be protected by a fuse to the manufacturer's recommended rating. The fuse is to be located in the base compartment of each mast. This compartment is to be protected by a close fitting vandal resistant door secured with a heavy duty minimum five lever security deadlock.

m. A separate ring main supply is to be used for perimeter lighting. Each fitting is to be on an individual spur from the ring main, and alternate lamps are to be on separate circuits.

n. A switch is to be fitted into the circuitry to permit the use of alternate luminaries for 50 % illumination if required.

o. Where switching of security lighting is performed automatically, switches are to be located at the security control room, not at the gate house. Back indication of the lighting state should be provided at the security control building, so that manual switching is immediately carried out when the automatic system fails.

2. When determining illuminance (Lux) or light output (Lumens), the following formula is to be used:

$$\frac{\text{Total output required}}{\text{Number of mounting points}} = \frac{A \times L \times 5}{N} = \text{Lumens per Lamp}$$

Where:
A = area to be illuminated
L = Illuminance in Lux
5 = Compensation factor for light loss between light source and ground.

Area is calculated by multiplying the length of the perimeter by the throw, throw being four times the mounting height of the luminary.

3. Area Lighting: Where security lighting is to be installed to illuminate an area, the following minimum requirements are to apply:

a. Under normal circumstances the minimum level of' illuminance at all points in the area is to be 30 lux.

b. At check points and other areas where vehicles are searched and documents are checked, the minimum illuminance is to be 50 lux.

c. Lamps are to be mounted on columns not less than 10 meters or more than 30 meters in height, except where specified to the contrary by the Secretariat to HCIS.

d. Lamps are to be high pressure sodium discharge lamps of' 400W, with a lighting design output of' not less than 45000 lumens. The restrike time is to be not more than one minute. The lamp is to reach full brilliance within five minutes. Lumen output beyond 2000 hours operation is not to decrease by more than 2% per 1000 hours operation.

e. Ballast and control gear are to be compatible with the power supply at the place of installation.

f. The lamp is to be mounted in a single body of heavy duty construction, of' a material which is corrosion resistant to elements present at the place of installation.

g. The reflector is to give an asymmetric beam.

h, The body is to fully protect the optical and electrical systems from outside elements.

i. The luminary is to be suitably earthed.

j. Each lamp is to be protected by a fuse to the manufacturer's recommended rating.

k. The system is to have robust fittings, and power is to be supplied to the mounting pole by armoured cables laid underground and rising within masts.

l. The ballast must reliably start the lamp in ambient temperatures down to -5°C.

4. Calculation of the number of lamps required is to be based on the formula:
Area (in m^2) x required illuminance (lux) x 4 = Total lamp lumens

5. Each column is to be on an individual spur from the ring main and alternate lamps are to be run on separate circuits.

6. In large open areas in excess of $100m^2$ lights are to be mounted on clusters.

7. Switching is to be in accordance with paragraph 1. o. above.

8. Lamp igniter distances are not to exceed manufacturer's specified maximum.

9. The selection of mounting height will depend on the area of the ground requiring illumination and the local environmental conditions, and is to be determined by consultation between the owner's representative and the Security/Safety Surveyors.

Unquote

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
3					

TES-S-101.16

Revision No. 0

SAFETY & SECURITY – CHECKPOINT LIGHTING

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1. SCOPE

This Standard refers to requirements for security checkpoint lighting system and is adopted verbatim from SAG SSD/14.

Quote

1. Checkpoint lighting is designed to highlight all vehicles and personnel who require access to the installation. Lighting levels at checkpoints will, therefore, be as high as possible to facilitate checking of personnel and documents. At checkpoints, the following standards will be followed:

   a. Lighting will be sufficient to allow people, vehicles, goods and documents to be checked thoroughly.

   b. Local floodlighting is to be installed to light up trucks and vans to ensure proper search.

   c. A light is to be installed under the canopy of the Gate House to facilitate the checking of drivers, of occupants of vehicles and their documents.

   d. The whole area of the vehicle lock, i.e. from the entrance gate to the end of the inner arm of the perimeter fence, is to be lit to a minimum level of 50 lux.

   e. The lighting in the Gate House is to be shielded and subdued by using dimmers.

2. A drawing showing the typical layout of a checkpoint and vehicle lock is available with the Secretariat.

3. Where mesh shields are provided, they are to be stored in the main Security Building for use only in an emergency. (See also SSD/17).

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
3					

TES-S-101.17

Revision No. 0

SAFETY & SECURITY – INTRUDER DETECTION SYSTEMS

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1. SCOPE

This Standard refers to requirements for intruder detection systems and is adopted verbatim from SAG SSD/15.

Quote

1. Where an Intruder Detection System (IDS) is installed, it must:

   a. Effectively cover the area and be highly resistant to tampering.

   b. Be acceptably free from unwanted (false) alarms.

   c. Be "fail safe" in that failure of any part of the system, whether accidentally or deliberately induced, must cause the alarm to operate.

   d. Continue to operate when any mains power it uses fails or is disconnected. Battery stand-by is to be used to meet this requirement and battery indication is necessary.

   e. The alarm indicator to be both audio and visual with a facility to accept and switch off the audio but leave the visual on the console.

   f. Where the system covers several zones of protection, each zone is to have its own visual signal but a common audio signal is acceptable.

   g. The alarm console to be incorporated in the system so that interference in the alarm console causes an alarm state.

   h. The alarm console to be mounted in a permanently manned location, for preference the security building with alternative arrangements in e.g. Main Control Room.

   i. Protection of the system is to be effected by using a fully monitored circuit.

2. The selection of an Intruder Detection System is a specialized matter. The advice of the Secretariat is to be sought at an early stage if the installation of an IDS system is projected.

Unquote

2. ADDITIONAL UNDERSTANDING

2.1 Generally this specification is a requirement for unmanned facilities, where facility warrants this type of protection.

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
3					

TES-S-101.18

Revision No. 0

SAFETY & SECURITY – AUTOMATIC CONTROL OF ENTRY (ACE) SYSTEMS

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1. SCOPE

This Standard refers to requirements for the Automatic Control of Entry (ACE) system into a facility and is adopted verbatim from SAG SSD/16.

Quote

1. It is emphasized at the outset that at the present time, no ACE system on its own is capable of:

   a. Providing security against surreptitious attack or forcible attack. Approved security locks should therefore be used whenever the system is not under supervision.

   b. Replacing human guard systems. The use of the system may take it possible to reduce the number of guards on control of access duties.

2. Because ACE systems are not proof against surreptitious attack, they depend to a degree of supervision either by a guard or a member of the staff.

3. The selection of an ACE system is a specialized matter. The advice of the Secretariat should be sought at an early stage if the installation of an ACE system is projected.

4. Basic requirements for an ACE system are:

   a. Acceptable to all users.

   b. Proof against forgery of key cards, tampering and manipulation.

   c. Supplied with overriding facilities for use in the event of an emergency.

   d. Supplied with an alarm facility that activates when a wrong card is inserted, or wrong key combination is selected.

   e. Designed to be used in conjunction with an IDS system.

   f. Fitted with security locks for use outside of working hours.

   g. Designed so that the operation of the push-button cannot be overlooked.

   h. "Fail Safe".

   i. Provided with stand-by power source with automatic cut-in.

   j. Capable of preventing repeated entry except where an entry is followed by an exit.

   k. So designed that the card key or push button codes can be changed quickly and easily.

l. Capable of restricting access to one person at a time.

Unquote

2. See also TES-S-101.25.

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
3					

TES-S-101.19

Revision No. 0

SAFETY & SECURITY – GATE HOUSE AND VEHICLE LOCK

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1. SCOPE

This Standard refers to requirements for the gate house and vehicle lock and is adopted verbatim from SAG SSD/17.

Quote

1. Where a gate house is constructed, it is to conform to the specifications as under. Under no circumstances are gate houses to be constructed or sited without prior reference to the Secretariat.

2. Gate houses are to be sited so as to completely control all vehicular and pedestrian traffic unless specified to the contrary by the Secretariat.

3. The fence for the vehicle lock associated with the gate house is to meet the same standards as that for the remainder of the perimeter fence.

4. Security lighting for the gate house and vehicle lock is to conform with that specified in SSD/13 and SSD/14.

5. Gate house construction is to be as follows:

   a. Base: The gate house is to be mounted on a base raised a minimum 500mm above the road surface.

   b. Lower Section: The lower section is to be constructed of concrete, minimum thickness 100mm, to a vertical height of 1 m above the raised surface of the base.

   c. Windows: Windows are to be of heavy tint glass, tilted from the vertical, to a height of 1 m above the concrete sill. Windows are to completely surround the building thus giving well round vision.

   d. Mesh: A shield is to be constructed of white painted wire mesh welded to a stout metal frame. This frame to fit exactly the dimension of the individual windows and is to be secured to the fabric of the building by lugs and padlocks.

   e. Roof: The roof of the gate house is to be constructed of lightweight non-flammable materials meeting the requirements of latest edition of NFPA 220 or better.

   f. Doors: There is to be only one door in the gate house situated at the site end of the building. It is to be outward opening, fitted with a single-sided lock with the key-way on the inside and to have a sentry viewer in the door. A bulk head light if the minimum 60W is to be mounted outside the door.

   g. External Lights. There is to be a minimum of two floodlights mounted on the leading edges of the gate house, one either side, to illuminate the area and to

highlight vehicles and foot traffic entering the vehicle lock. The lights are to provide a minimum 50 lux. (See also SSD/14).

h. Communications: The gate house is to be provided with telephone communications from the PABX and also a direct line to the security building or main control room. Where radio is installed, the gate house is to be provided with a set for normal security channel usage.

i. Barriers: Two electrically operated barriers are to be installed one each side of the gate house, to control vehicular traffic. Additionally, the barriers are to be fitted with manual overdrive to allow them to be operated in the event of a loss of power. To further control vehicular traffic and reduce the risk of collision of vehicles with the barrier, the 'IN' side of the vehicle lock is to be provided with a minimum three rumble strips laid at not more than 1m centres.

j. Internal Lighting: The interior of the gate house is to be fitted with dimmed lighting shielded from view. Ceiling and walls are to be painted with subdued colours.

k. Checkpoints: There shall be two windows designated as checkpoints, one each side of the building. These windows are to be designed so as to prevent unauthorized access from the outside. The lower 1/3 of the window is to be fitted with a sliding section sufficient to allow the passage of documents and passes for scrutiny by the guard on duty inside the gate house.

6. Where there is a requirement to install alarm consoles, security lighting switches or other control room devices in the gate house as opposed to the security building, then the Secretariat must be consulted before any planning decision is taken.

Unquote

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
3					

TES-S-101.20

Revision No. 0

SAFETY & SECURITY – FIRE FIGHTING EQUIPMENT

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1. SCOPE

This Standard refers to requirements for fire fighting equipment and is adopted verbatim from SAG SSD/18.

Quote

1. This specification covers the general requirements for the design, construction, material selection and the testing of fire fighting equipment for petroleum refineries and related offsite facilities.

2. The latest edition of the following National Fire Codes will be referred to and used as a minimum guide when ordering, building, installing, maintaining and/or altering fire equipment and appliances.

a.	NFPA 1	:	Uniform Fire Code.
b.	NFPA 10	:	Standard for Portable Fire Extinguishers.
c.	NFPA 11	:	Standard for Low, Medium and High Expansion Foam.
d.	NFPA 11A	:	Standard for Medium and High Expansion Foam.
e.	NFPA 12	:	Standard on Carbon Dioxide Extinguishing System
f.	NFPA 13	:	Standards for the installation of sprinkler systems, including appendices.
g.	NFPA 14	:	Standard for the Installation of Standpipe and Hose System.
h.	NFPA 15	:	Standard for Water Spray Fixed System for Fire Protection.
i.	NFPA 16	:	Standard for the Installation of Foam-Water Sprinkler and Foam-Water Spray Systems.
j.	NFPA 17	:	Standard for Dry Chemical Extinguishing System
k.	NFPA 17A	:	Standard for Wet Chemical Extinguishing System
l.	NFPA 20	:	Standard for the installation of Stationary Pumps for Fire Protection.
m.	NFPA 22	:	Standard for Water Tanks for Private Fire Protection
n.	NFPA 24	:	Standard for the Installation of Private Fire Service Mains and their Appurtenances.

o.	NFPA 25	:	Standard for Inspection, Testing and Maintenance of Water Base Fire Protection System
p.	NFPA 30	:	Flammable and Combustible Liquids Code.
q.	NFPA 58	:	Liquefied Petroleum Gas Code.
r.	NFPA 59A	:	Standard for the Production, Storage and Handling of Liquefied Natural Gas (LNG).
s.	NFPA 68	:	Standard for Explosion Protection by Deflagration Venting.
t.	NFPA 70	:	National Electrical Code
u.	NFPA 72	:	National Fire Alarm Code
v.	NFPA 75	:	Standard for Protection of Information Technology Equipment
w.	NFPA 90A	:	Standard for Installation of Air Conditioning and Ventilating Systems
x.	NFPA 101	:	Life Safety Code
y.	NFPA 214	:	Standard on Water-Cooling Towers.
z.	NFPA 230	:	Standard for the Fire Protection of Storage
aa.	NFPA 291	:	Recommended Practice for Flow Testing and Marking of Hydrants.
bb.	NFPA 307	:	Standard for the Construction and Fire Protection of Marine Terminals, Piers and Wharves.
cc.	NFPA 329	:	Recommended Practice for Handling Releases of Flammable and Combustible Liquids and Gasses.
dd.	NFPA 385	:	Standard for Tank Vehicles for Flammable and Combustible liquids.
ee.	NFPA 750	:	Standard on Water Mist Fire Protection Systems
ff.	NFPA 780	:	Standard for the Installation of Lightning Protection Systems.
gg.	NFPA 1901	:	Standard for Automotive fire apparatus.
hh.	NFPA 1961	:	Standard on Fire Hose.

ii. NFPA 1963 : Standard for Fire Hose connections.

jj NFPA 2001 : Standard on Clean Agent Fire Extinguishing System

3. Roads surrounding the area at risk will be graded to fully withstand the maximum weight of fire fighting vehicles and associated outriggers.

4. Compatibility of equipment must be maintained when purchasing new, or modifying existing equipment.

Unquote

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
3					

TES-S-101.21

Revision No. 0

SAFETY & SECURITY – WORK PERMIT PROCEDURES

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1. SCOPE

This Standard refers to requirements for work permit procedures and is adopted verbatim from SAG SSD/19.

Quote

1. The following work permits only are authorized:

   a. Hot Work

   b. Cold Work

   c. Entry

   For layout, see specimens attached.

2. Under no circumstances will an ENTRY permit be issued unless accompanied by a HOT or COLD work permit.

3. Work permits will clearly delineate the responsibilities of signatories.

4. Work permits will only be valid for the duration of one shift only.

   Note: A list of staff authorized to issue permits will be maintained and strictly observed by all personnel.

5. It is essential that all personnel authorized to issue permits are fully trained within their field of operations.

Unquote

HOT WORK PERMIT

DATE:_____ SERIAL NO: _____

1. Location of Work Site:

   Work to be Carried Out:

   Equipment/Vehicles to be used:

---

2. Associated Permits:

   Entry Permit No:_____ Electrical Permit No:_____

   Other Permit No:_____ Key No:_____

---

3. Isolation: Equipment to be isolated by:

   a. Valve Closure_____ c. Line Removal_____

   b. Blanks/Spades_____ d. Electrically_____

---

4. Conditions of Work: (Delete as necessary)

   a. Inform Area Supervisor before work commences.
   b. Area Supervisor to be on site when work commences.
   c. Conditions of work to be approved by Area Superintendent.
   d. Area/Equipment to be cleared of combustible materials.
   e. Gas concentration to be continuously monitored.
   f. Fire extinguishers to be sited. _____(State type)
   g. Scaffolding required.
   h. Safe means of access required:
   i. Safety Inspector to Stand-by
   j. Fireman to Stand-by
   k. Additional items listed below:-

---

5. Gas test required: Yes/No_____ Frequency: Every_____Hour or less

   a. Flammable. Result_____ Time_____ Signature_____
   b. Toxic. Result_____ Time_____ Signature_____

---

6. I have personally checked the above site and conditions. Permission is granted for Hot Work to commence.

Time From:_____To_____Date: _____

Signature:_____Area Authority.

---

7. Safety Department has been consulted and agree with the above conditions.

Further recommendations:

Signature: _____Safety Inspector.

---

8. I understand the conditions and only work permitted will be attempted. All conditions will be observed.

Signature: _____Person in Charge of Work.

---

9. Work Completed.

Time: _____ Date: _____

Signature: _____Person in Charge of Work.

---

10. Work Accepted.

Time: _____ Date: _____

Signature: _____Area Authority.

Distribution:- Original - Person in Charge of Work.
1st Copy - Area Authority
2nd Copy - Safety Department.
3rd Copy - Stays in Book.

COLD WORK PERMIT

DATE:_____ SERIAL NO: _____

1. Location of Work Site:

Work to be Carried Out:

Equipment/Vehicles to be used:

---

2. Associated Permits:

Entry Permit No:_____ Electrical Permit No:_____

Key No:_____

---

3. Isolation: Equipment to be isolated by:

a. Valve Closure_____ c. Line Removal_____

b. Blanks/Spades_____ d. Electrically_____

---

4. Conditions of Work: (Delete as necessary)

a. Inform Area Supervisor before work commences.
b. Conditions of work to be approved by Area Superintendent.
c. Area/Equipment to be cleared of combustible materials.
d. Gas concentration to be continuously monitored.
e. Scaffolding required.
f. Safe means of access required:
g. Roads to be closed
h. Safety Inspector to Stand-by
i. Fireman to Stand-by
j. Additional items listed below:-

---

5. Gas test required: Yes/No_____ Frequency: Every_____Hour or less

a. Flammable. Result_____ Time_____ Signature_____

b. Toxic. Result_____ Time_____ Signature_____

---

6. I have personally checked the above site and conditions. Permission is granted for Cold Work to commence.

Time From:_____To_____Date: _____

Signature:_____Area Authority.

---

7. Safety Department has been consulted and agree with the above conditions.

Further recommendations:

Signature: _____Safety Inspector.

---

8. I understand the conditions and only work permitted will be attempted. All conditions will be observed.

Signature: _____Person in Charge of Work.

---

9. Work Completed.

Time: _____ Date: _____

Signature: _____Person in Charge of Work.

---

10. Work Accepted.

Time: _____ Date: _____

Signature: _____Area Authority.

Distribution:- Original - Person in Charge of Work.
1st Copy - Area Authority
2nd Copy - Safety Department.
3rd Copy - Stays in Book.

ENRTY PERMIT

DATE:_____ SERIAL NO: _____

1. Location of Work Site:

Work to be Carried Out:

Equipment/Vehicles to be used:

---

2. Associated Permits:

Hot Work Permit No:_____Electrical Isolation Permit No:_____

Cold Work Permit No:_____ Key No:_____

---

3. Isolation: Equipment to be isolated by:

a. Blanks_____ c. Line Removal_____

b. Spades_____ d. Electrically_____

---

4. Conditions of Work: (Delete as necessary)

a. Inform Area Supervisor before Entry and / or Re-entry.
b. Conditions of entry to be approved by Area Superintendent.
c. Area/Equipment to be cleared of combustible materials.
d. Gas concentration to be continuously monitored.
e. Vessel/Area to be gas free
   - Flammable Gas.
   - Toxic Gas. State type _____
   - Oxygen deficiency.
f. Protective clothing / equipment required:
   - Air supplied breathing apparatus.
   - Life line harness.
   - PVC Overall Suit.

   Additional equipment:

g. Safety Inspector to Stand-by
h. Fireman to Stand-by

---

5. Gas test required: Yes/No_____ Frequency: Every_____Hour or less
Before Re-entry: Yes/No_____
a. Flammable gas reading: _____Time_____Signature_____
b. Toxic gas reading: _____Time_____Signature_____
c. Oxygen content: _____Time_____Signature_____

6. I have personally checked the above site and conditions. Permission is granted for entry.

Time From:_____To_____Date: _____

Signature:_____Area Authority.

---

7. Safety Department has been consulted and agree with the above conditions.

Further recommendations:

Signature: _____Safety Inspector.

---

8. I understand the conditions and only work permitted will be attempted. All conditions will be observed.

Signature: _____Person in Charge of Work.

---

9. Work Completed.

Time: _____ Date: _____

Signature: _____Person in Charge of Work.

---

10. Work Accepted.

Time: _____ Date: _____

Signature: _____Area Authority.

Distribution:- Original - Person in Charge of Work.
1st Copy - Area Authority
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REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.22

Revision No. 0

SAFETY & SECURITY – EXPANDED METAL MESH - DIMENSIONS

PROPRIETARY INFORMATION

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1. SCOPE

This Standard refers to dimensions of expanded metal mesh for the caging requirements of TES-S-101.12 and adopted verbatim from SAG SSD/20.

Quote

Mesh Size SW (mm)	LW (mm)	% of Open Area		Size Of Strand		Weight (kg/m^2)
		Normal	Maximum	Width	Thickness	
12.70	30.48	77	81	1.19	1.20	1.90
12.70	30.48	77	83	1.63	0.95	2.05
12.70	30.48	68	78	2.06	1.20	3.32
12.70	30.48	62	76	2.36	1.20	3.81
12.70	30.48	67	75	1.98	1.58	4.20
12.70	30.48	59	71	2.36	1.58	4.98
12.70	38.10	73	86	2.31	1.20	2.59
12.70	38.10	74	83	2.31	1.58	3.37
12.70	38.10	62	76	3.10	1.58	4.54

Note: This matrix to be read in conjunction with the specifications outlined in SSD/10.

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REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.23

Revision No. 0

SAFETY & SECURITY – THE PROTECTION OF DIESEL ENGINES

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1. SCOPE

This Standard refers to requirements for diesel engines operating in hazardous areas and is adopted verbatim from SAG SSD/21.

Quote

1. All diesel engines, new or existing, mobile or fixed, must comply with the following specific standards when being operated in hazardous areas. (Division / Zone 1 & 2 areas).

2. The following measures will be carried out by the manufacturer before new diesel engines are brought into operation inside hazardous areas.

   a. All external. metallic parts e.g. fan blades which may cause sparking by contact with other parts, either when running normally or when malfunctioning, are to be manufactured from non-sparking materials.

   b. All driving belts are to be manufactured from antistatic materials.

   c. The engine may not be started by a battery powered electric starter motor. Air, inertia, spring or mains powered electric flameproof motor starters are to be used.

   d. An approved type of spark arrestor is to be fitted downstream of the exhaust manifold.

      NOTE:

      (1) The approved spark arrestor may either be of the dry cyclone type, the baffle type or the water bath gas conditioner box type.

      (2) If a gas conditioner box is fitted, it is to be constructed of non-corroding materials and suitably baffled to prevent 'water carry-over' with the exhaust gas and ensure thorough mixing of gas and water. In addition, it is to be provided with a water storage tank and devices to maintain the water in the box at the correct level and sound an audible alarm or automatically stop the engine if the water falls below the level.

   e. Approved type of flame traps are to be fitted to the air intake, downstream of the air filter (see Note 3 below). Flame traps are to be manufactured from non-corroding materials and are to be capable of easy dismantling for cleaning and inspection purposes.

   f. The surface temperature of any part of a diesel engine is not to exceed 250º C in maximum load conditions.

g. An approved device is to be fitted into the air intake which will automatically close the air inlet and stop the engine if it exceeds its normal governed speed. The closing of the air inlet must be sufficiently tight to stop the engine whether running on normal fuel or on induced flammable vapors. Additionally, a manual closing system is to be provided. In cases where the engine is to be incorporated in mobile equipment, this manual control is to be operated from the driver's cabin.

h. The fuel pump and cam shall be so designed that a reverse running of the engine is not possible.

i. The length of the flame path through or across any joint in the air intake system is to be not less than 13mm.

j. Suitable jointing material is to be interposed between all joint faces in the air intake and exhaust systems to ensure tight joints.

k. Where valve spindles pass through the walls of any component in the induction system, the diametrical clearance is not to exceed 127mm for an axial length of 25mm unless end caps are fitted.

l. No screw, stud or bolt hole is to pass through the walls of any component of the inlet system.

m. All cylinder head and crank case relief valves and breathers are to be connected to an approved flame trap as in (e) above.

n. The exhaust outlet to atmosphere is to be arranged as high above the ground level as possible.

o. All electrical equipment is to be certified by a statutory body acceptable to the Secretariat to the High Commission for Industrial Security.

p. Decompression systems directly exhausting to atmosphere are not to be provided.

q. All mobile equipment, powered by a diesel engine and intended for use within a Hazardous Area must be distinctively marked.

3. Modification to Existing Equipment

For all existing diesel engines and associated driven equipment intended for use within Hazardous Areas, the following modifications are required:

a. Provision of an approved type of spark arrestor (see d. above).

b. Provision of an approved type of flame trap (see e. above).

c. Incorporation of an approved type of over-running shut-down valve (see g. above).

d. Check exhaust manifold and exhaust pipe surface temperatures with Templesticks (Thermocolor-Faber-Castell) and if those are indicated to be above 250ºC, these components should be protected and insulated by fiberglass lagging with aluminum or steel cladding.

e. All driving belts are to be manufactured from antistatic material (see b. above).

Note (3) When stationary diesel engines are installed so that the air intake is in a 'Safe Area', the precaution under 2 e. above, so far as it concerns the air intake system and 2 g. above concerning the over-running shut-down device, need not be applied.

Note (4) Where stationary diesel engines are installed so that the exhaust gases are discharged in a 'Safe Area' the precautions under 2 d. above so far as they concern the exhaust system need not be applied.

Note (5) In general, mechanical equipment should be fitted throughout and electrical equipment avoided.

Note (6) Starting of all existing diesel engines with non-flame proof electric starting equipment is to be effected only under permit.

Note (7) When existing equipment is modified for use in Hazardous Areas as in 3 above, measures should be taken to ensure that use is not made of vehicle lighting or auxiliary electrical systems while operating in a hazardous area.

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REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
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TES-S-101.24

Revision No. 0

SAFETY & SECURITY – GASOLINE FILLING STATIONS

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1. SCOPE

This Standard refers to requirements for gasoline filling stations and is adopted verbatim from SAG SSD/22.

Quote

1. Gasoline Filling Station on Industrial Sites are to be built to the following specifications:

a. Forecourt

(1) The area formed by a cone, the apex of which is the top of the pump enclosure, the base at forecourt level with a radius of 2.43 m from the pump will be classified as Division/Zone 2.

(2) Any electrical fitting installed within 22.86 cm of the forecourt level in a Division/Zone 2 area, up to a limit of 2.43 m from the pump, is to be of a design acceptable for Division/Zone 1 application.

b. Pump Enclosure

(1) The pump enclosure will be classified as Division/ Zone 1.

c. Canopy-Kiosk/Office

(1) The area within a Kiosk/Office located on a pump island is to be classified Division/Zone 2 up to a height of 1.22m above floor level.

(2) Electrical equipment other than cables will not be installed within 22.86cm of floor level.

(3) Light fittings installed above 1.22m from floor level will be of the enclosed type.

d. Gasoline Tanks-Vent Pipes

(1) An area within 1.52m, in any direction of the vent discharge pipe outlet will be classified as Division / Zone 1. Elsewhere, the area within 1.52m of the vent riser pipe will be classified as Division/Zone 2.

(2) The open end of the pipe will discharge upwards into the open air (out of reach) and be at least 3.65m above ground level. The pipe discharge point will not be located within 1.52m of any window or other openings.

(3) Vent pipes will not be installed within 1.52m of any boundary/line or fence, except where it is a solid wall extending for 1.52m beyond the

vent discharge point in every direction, including down to ground level; this wall will be considered as the limit of the classified area.

2. Gasoline Pumps

1. The number of pumps on each site will be kept to a minimum, capable with efficiency and the smooth functioning of the site.

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No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
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TES-S-101.25

Revision No. 0

SAFETY & SECURITY – AUTOMATIC CONTROL OF ENTRY

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1. SCOPE

In addition to TES-S-101.18, this standard refers to requirements for the Automatic Control of Entry (ACE) into a facility and is adopted verbatim from SAG SSD/23.

Quote

1. This SSD is to be read in conjunction with SSD/16, the provisions of which, particularly with regard to purchase of ACE systems and the need to consult with the Secretariat, still apply.

DESIGN AND FUNCTION

2. The system must allow exit and entry to or from a protected area by means of individually coded pass-cards together with a memorized personal code number.

3. Each access point to a protected area is to be controlled by a Card-reader to be mounted in close proximity to the door or turnstile and on approval of a card and personal code the Card reader to operate an electrical release mechanism thus allowing the cardholder to pass.

4. Each card is to contain a unique code so that lost or stolen cards can be blocked out of the system by means of a Blocking Unit. Where a single point system is in use, the Blocking Unit to be a ten way insert fitted within the Card reader. Where a multipoint system is in use, the Centralized Blocking Unit to be a 24 way insert.

5. A Conditional Insert to be used to program groups of cards to allow access only during certain times of the day or on particular days of the week. The validity of a particular group of cards is to be controlled by a contact on an External Time Clock.

6. A clear text printer is to be included in the system to provide a continuous record of each cardholder's entry or exit together with the number of access points used, the time and the date. Attempted entries using blocked cards are also to be recorded.

7. Where both entry and exit are controlled by Card readers, an electronic memory is to be installed and read-out facility to be added to the system whose main function will be to indicate immediately:

   a. Whether a particular cardholder is within the protected area.

   b. The prevention of one card being used to admit two or more people.

EQUIPMENT

8. The Card reader to be a self-contained unit of modular design. It is to use R.F. field technique for reading the card data which will normally carry at least 25 bits of information.

9. In addition to card information, there is to be a personal code of not less than four digits which the holder of the card must enter on the Card reader keyboard.

10. The Card reader unit to contain the following items:

    a. A Keyboard with a minimum of nine push-buttons for entering on the Card reader keyboard.

    b. A slot in which the card can be inserted to be read by the Card reader.

    c. A system of lights and words identifying actions to be taken / completed.

        (1) Amber light and the word "CODE".

        (2) Green light and the word "OPEN".

        (3) Red light and the word "REJECT".

11. Within the Card reader, the following equipment to be included:

    a. A "motherboard" containing the Card reader control circuitry, a connector strip for all external wiring and four edge-connectors to accommodate the plug-in inserts are to be used.

    b. A programmed Conditional Insert that enables acceptance of approved card groups only.

    c. A scrambling Insert for personal codes. This insert to have a minimum five variations.

12. The equipment to be so designed that it will only react to cards that contain readable elements that are part of that system.

<u>OPERATIONS</u>

13. The equipment to operate in the following manner:

    a. On insertion of the card the Card reader to react by lighting up one of the three lamps identified above.

    b. On identifying the Code instruction the Cardholder to dial his personal code by means of the push-button.

    c. The correct dialing of the personal code to activate the OPEN lamp and spring the lock open for access.

    d. In the event of a mis-dial or of the wrong number being dialed the REJECT lamp to light up.

e. The door to remain open long enough for the Card Holder to pass through, but the time to be limited, see below.

f. The Card reader to be equipped with an alarm status circuit to indicate locally or remotely:

(1) The door has been opened by means other than a card.

(2) The door has been opened with a proper card but has not been closed within a stipulated time. For ease of reference this time-band should be not less than 5 seconds or more than 30 seconds.

(3) Incorrect personal codes have been entered more than three times.

14. The Card reader is to be protected with a tamper alarm which can be connected to an independent intruder alarm system or any other alarm circuit in use in the building / installation.

15. The system to incorporate an additional relay to shunt the intruder alarm contact when a correct card had been presented and the correct code dialed.

16. Provisions to be made for the inclusion of override switches to be used if there is a requirement to release the door lock from a remote point or to operate the release device by means of an exit push button on the inside of the controlled entrance.

17. The Clear Text Recorder is to be utilized to provide up-to-the-minute information on who goes through what door and when.

18. Mains operated Time Switches are to be incorporated in the system so that certain groups of cards can be prevented from entering during certain periods.

19. Where multi-point system is in use the Card readers are to be fitted with an additional plug-in Transmitter Insert which is to interface with the Card reader with the common Central Equipment.

20. For centralized multi-point access systems a control unit/scanner, to which all card readers are connected, is to be installed. Where there are more than 32 card readers, it will be necessary to install two or more scanners in parallel.

21. The Scanner is to operate as an electronic stop-selector which is to continuously scan the output circuits of the Card readers and connect one Card reader at a time to the common recorder and Blocking Unit.

22. The Blocking Unit is to be secured against tampering with a stout, robust steel case capable of being locked and secured at all times except for authorized access.

23. Clear Text Recorders are to be powered from a 24V D.C. supply and print at a minimum speed of one line per second each line having 18 columns.

24. The printout to show the following data:

    a. Month/Day

    b. Hour, Minute

    c. Door Number

    d. Identify Number

25. Each printer interface is to be fitted with a built-in electronic clock which automatically supplies the Recorder with the following information:

    a. Date

    b. Hour

    c. Minute

26. The Recorder to be fitted with a push-button which, when activated will print-out the date, hour and minute on one line. This will be used for checking that the Recorder is operational and also that the clock is giving the correct information.

27. The Recorder is to be fitted with override circuits. The circuits are to be capable of bypassing the recording function if the recorder jams or there is a mains power failure. These conditions to be indicated at a remote point by a signaling circuit in the Recorder.

28. The Recorder must place on record all REJECT attempts to gain entry. The information is to be printed in RED so that it is immediately apparent to the monitoring authority.

29. Where very high security is required the Memory Unit is to be installed as standard equipment. The following applications are to be followed when a Memory Unit is installed:

    a. Both Entry and Exit are to be controlled by Card reader.

    b. A Card cannot be used to gain access a second time unless it has first been used to exit the area.

    c. The Memory Unit is capable of handling at least 4000 personal codes and 15 entry/exit points.

30. Memory Units are to be fitted with stand-by batteries. However, it is essential to realize that a loss of power, caused by either a circuit or a service engineer's action, will erase the information stored in the unit. If such an incident occurs it will be necessary to redial all the information.

31. The card in use is to be fully laminated to the "data collection" size. It is to contain invisible and passive code elements.

32. Codes on the card are to be arranged as follows:

    a. Octal code from which the personal code is derived.

    b. System code from which the Card reader identifies the particular card for the system.

33. A separate list is to be provided with information about each individual card with its personal code since there is to be no indication on the card of the relevant code. This list will need to be given a high degree security within the system.

34. The cards to be manufactured according to international standards for data collection cards. The code elements to be placed between two pieces of high quality plastic material and RF welded together.

35. Cards are to be printed with a serial number which in no way is to be connected with the octal or personal code.

36. A commuter list is to be provided with the equipment showing:

    a. Octal serial number

    b. Octal number

    c. Personal Code Number (5 different codes)

    d. Blocking code.

RELEASE SERVICES

37. The lock in use is to be CHUBB Castle or better fitted with an electronic striking plate type SOLIDLAS 514 BSE 24v (125mAO). The plate is to be fail-safe designed for continuous use and have a built-in control contact to ensure the door is not only closed but also locked.

38. Where circumstances dictate that the striking plate be locked when energized then such information must be stipulated when ordering the system since this condition requires the adjustment of the Card reader.

USAGE

39. The ACE system outlined above is to be used only in buildings already sited within a security compound or other security area. It is not to be used for control of access at the main gate to the security area, compound or installation.

Unquote

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
3					

TES-S-101.26

Revision No. 0

SAFETY & SECURITY – TURNSTILES FOR ENTRY CONTROL

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1. SCOPE

In addition to TES-S-101.18 and TES-S-101.25, this Standard refers to requirements for Turnstiles used in entry control and is adopted verbatim from SAG SSD/24.

Quote

TURNSTILES USED IN CONTROL ON ENTRY

1. This SSD is to be read in conjunction with SSD/16 and SSD/23.

2. Turnstiles will be installed only with the written permission of the Secretariat. Where permission is granted, the turnstile is to be used as part of an Automatic Control of Entry System.

3. The purpose of the turnstile shall be to stop un-authorized entry while permitting authorized entry with maximum convenience.

4. Turnstile shall be of rugged construction to withstand the constant use expected in an industrial environment.

5. The Turnstile shall be full height, that is, not less than 2 meters clear walkway. The minimum walkway width shall be not less than 500 mm and the maximum not more than 600 mm.

6. The central rotor shall have horizontal bars set at 90º to each other set. The bars shall be set not more than 155 mm apart.

7. The turnstile shall be manufactured from stainless steel or similar approved material to ensure long life with the minimum of maintenance. Where necessary, suitable toughened glass or similar transparent material shall be used to ensure the minimum claustrophobic effect.

8. Stators will be provided to confine access via the walkway.

9. The ends of the rotor bars shall not project unprotected when the turnstile is at rest. It shall not be possible for the turnstile to lock when a person is passing through.

10. Turnstile should be self-supporting by bolting to the floor with raw bolts or similar, it should not be necessary to excavate the floor to install them. Ideally the units should be delivered to the site in modular form.

11. The central rotor shall be controlled by an electro-mechanical unit mounted above the walkway area or the turnstile. The control device will be capable of locking or releasing the rotor in either direction.

12. A form of "anti back-up" device shall be used such that once the rotor has been released from the locked position and moved through 25º it will not be possible to

return it to its original position. This "Back-up" will be progressive to stop "rocking" of the motor.

13. Subject to the requirements of the User, either a fail-safe or a fail-locked facility under loss of power will be incorporated. The control device will be capable of receiving instructions from card reader devices or similar via an interface unit which will also provide facilities for interlock and alarms. The control system will operate on 24v DC and a suitable transformer or rectifier must be provided.

14. The unit shall, with the addition of minimum environmental control, be capable of installation in the open.

15. Provision shall be made in the capability of the interface and power supply to provide walkway illumination and flow control lights.

16. In areas where space is a prime consideration, double units shall be installed with all the facilities offered by the single unit but with the ability to save space by inter-meshing the two rotors.

17. The flow rate for a controlled turnstile is to be not less than ten persons per minute based on six seconds per controlled entry. This period includes the operation of the card reader and passing through the turnstile when it is released.

18. Where on-push-button controlled egress is applied to a turnstile the flow rate shall be not less than 15 persons per minute.

19. The basic requirements for an access controlled turnstile are:

    a. A card reader on the entry side of the turnstile and a Card reader or pushbutton the exit side depending on the control system specified for exit.

    b. Separate electronically motivated mechanical release for each direction of rotation.

    c. Mechanically operated contacts which signal the positional information of the turnstile to the Card reader.

    d. Key release of the turnstile mechanism to allow access in the event of a main failure. Access in this situation is to be supervised by Security personnel.

    e. A security lock fitted to the mechanism cover to prevent unauthorized tampering and tamper alarm contacts.

<u>Unquote</u>

2. This specification should be used in conjunction with TES-S-101.18 and TES-S-101.25.

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
3					

TES-S-101.27

Revision No. 0

SAFETY & SECURITY – INTERNAL CLEANING OF STORAGE TANKS

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1. SCOPE

This Standard refers for the internal cleaning of storage tanks that have contained hydrocarbon liquids and is adopted verbatim from SAG SSD/25.

Quote

1. The following procedures are to be strictly adhered to, before and during the Internal Cleaning of Storage Tanks that have contained Hydrocarbon liquids.

2. The Procedures

a. Contents of tank to be pumped out where possible by normal means, to the lowest level.

b. Cold work and/or Entry Permit must be issued and the contents thereon complied with. (SSD/19 refer).

c. One 12 kg Dry-Powder or B.C.F. fire extinguisher will be positioned by every entrance to the tank.

d. All pipe work connected to the tank must be spaded off, unless a valve or spool piece is fitted into the pipe work immediately before the tank flange, then the valve and/or spool piece will be removed.

e. An Air-Mover will then be securely bolted on to one of the tank manholes or pipe flange.

f. Ensure that the earthing cable of the tank is in good order and is properly grounded.

g. Connect Air-Mover to compressor or steam unit and commence degassing tank.

h. Take combustible and toxic gas tests from outside of the tank (unless wearing Compressed Air Self-Contained Breathing Apparatus).

i. Ensure that there is sufficient oxygen content in the tank, for personnel to work without breathing apparatus.

j. If tank has held sour crude, a water spray must be used on the inside walls of the tank continuously until all scale has been removed.

k. All personnel, before entering the tank, will don a safety harness and have a life-line attached.

l. Any machine, vehicle or other apparatus that is fitted with or driven by a Diesel Engine, must conform to SSD/21.

m. Only 'Certified' Flame-Proof lighting will be used in the area.

n. Only non-ferrous tools will be used for cleaning the inside of tanks, until the tank and surrounding area has been certified 'Gas Free'.

o. Only intrinsically safe equipment will be used within the diked area of the tank.

p. All residue of tank to be taken outside of diked area.

q. At all times when there are personnel working within a tank, a stand-by will be positioned outside with a rescue line and a set of breathing apparatus, at the ready.

Unquote

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
3					

TES-S-101.28

Revision No. 0

SAFETY & SECURITY – BLAST RESISTANT CONTROL BUILDINGS

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1. SCOPE

This Standard refers to the requirements for blast resistant control buildings and is adopted verbatim from SAG SSD/26.

Quote

1. New single and multi-unit control building located in refineries, petrochemical plants, Gas & Oil Supply Points or in other similar blast-hazardous environments, shall be blast resistant, single storey and without windows.

2. A minimum clearance of 35 meters is required between the control room building and equipment or vessels handling flammable liquids and/or gases, or buildings housing explosive materials.

3. Non-related offices, dining and recreational rooms will be separate from the control building.

4. The control building will be designed to prevent the entry of flammable liquids and vapors.

5. The source of air for purging control building shall be free of hazardous concentrations of flammable vapors and gases, contaminants and any other foreign matter. The source of air shall be determined from the nature of the process and the physical layout.

6. Ducting shall be constructed of non-combustible materials. The fan suction line shall be free of leaks and given suitable protection from mechanical damage and corrosion to prevent concentrations of flammable from being admitted to the control building.

7. The air system shall be design to provide positive pressure ventilation for all areas for the building. The positive pressure air system shall:

   a. Be capable of maintaining a pressure of at least 0.1 inch of water in the control-room with all openings closed.

   b. Be capable of providing a minimum outward velocity of 60 feet per minute through all openings. All doors capable of being opened shall be considered open, and an allowance for other openings shall be included.

8. It shall be acceptable for the positive pressure air system to include heating, ventilation and air-conditioning equipment, plus any auxiliary equipment found necessary to comply with the above.

9. A minimum number of doors should be provided so that positive pressures can be maintained, while at the same time, the number of doors should be adequate for safe exit.

10. The control room shall be electrically isolated by space and a wall, from electrical and radio switchgear rooms. Shielding from this equipment will be built into this dividing wall.

11. Process, chemical or hydrocarbon pipe work will not be piped into, or through the control building.

12. No process or operational handling and disposal of liquids shall take place, or be located over or under the control building.

13. Manufacturers of blast resistant doors, supplied for fitting into control buildings, must furnish a warranty as to their suitability.

Control Building Sizing

14. The length and arrangement of the control boards, as well as access to the back of the boards, shall determine the size of the control room. An example of a control room building, suitable for housing a control board 32 feet long is shown in attachment 26 - 1A - 2.

Special Design Loads

15. New control buildings located at 35 meter minimum from a potential blast source shall be designed to withstand the maximum combination of dead loads resulting from anyone of the following:

    a. Overpressure equivalent to 3 psi static pressure acting on all exterior surfaces.

    b. Suction overpressure equivalent to 1 psi static pressure acting on all external surfaces.

    c. Effective wind velocity pressure of 75 PSF on flat surfaces.

16. New control buildings located more than 100 feet from a blast source shall be designed in accordance with the following table:

DISTANCE VERSES DESIGN PRESSURES

Distance from Control Room to Process Equipment (Feet)	Over Pressure (PSI)	Suction Overpressure (PSI)	Wind Velocity Pressure (PSF)
100 to 150	3	1	75
150 to 200	1.5	0.5	38
200 to 250	1.0	0.33	25
250 to 500	0.5	0.17	*
Over 500	*	*	*

* No special design loads.

17. The following permissible loads stresses shall be used when designing to the special design loads given in the above table:

    a. For overpressure and suction overpressure, as noted in paragraph 15, item (a) and (b), normal allowable stresses shall be used, except for shear, where the stress shall be limited to a maximum of 75% of normal.

    b. For wind velocity pressure, as noted in paragraph 15, item (c), 133 and 1/3 per cent of the normal allowable stress (due to the combination of wind and dead loads) shall be used .

Construction Standards

18. Construction standards for building elements (foundations, structures and enclosures) are determined by the foregoing design load data. A typical wall section for a reinforced concrete, blast resistant building is shown in attachment 26 - 2A - 2.

19. Openings in the building enclosure shall be minimized and arranged to avoid or be shielded from direct blast pressures. Recesses and blast shields, as shown in 26 – 1A - 2 should be used only on the walls remote from the most likely direction of the potential blast source. Openings larger than 8 inches shall be investigated for the effect of shock resistance. Openings that could cause pressures on top surfaces of hung ceilings shall not be used.

20. External doors, louvers and similar items, together with their frames, shall be capable of withstanding the overpressures. Glass panels shall not be used in these doors. Two means (doors) of ingress and egress from the control room, located remote from each other and where possible not in opposite walls, shall be provided for personnel. Openings shall be sized for maneuvering factory-fabricated control boards into or out of the control room. External door frames shall be set in a rabbet on all sides, including the bottom, to prevent the door and frame from being blown in. Doors shall open outwards only, and shall be provided with closers.

Detailed Design

21. The following design requirements will be incorporated in the building:

    a. Walls shall be cast-in-place reinforced concrete of load bearing type, designed to resist bending and transit horizontal sheer. Pre-cast concrete panels may be used but shall be either mechanically bonded to cast-in-place reinforced concrete columns or provided with built-in, load transmitting steel plates or angles so that the panels can be welded in place.

    b. Walls shall be anchored to foundations and concrete roof slabs with steel reinforcing bars to provide full moment connections.

c. Roof framing shall be provided with adequate bearing and good anchorage to the supporting walls, roof form decking if used, should be welded to the top member of the roof frame.

d. Loads on roofs and walls shall be distributed in two directions, where possible.

e. Stress reversals at each member shall be considered, and a minimum 20% strength reversal shall be provided.

f. Reinforced concrete members shall have good ductile properties. Therefore, tension steel shall be limited to 2% of the concrete area in order to prevent any failure in tension. To allow for stress reversal, reinforcing steel shall be provided for both concrete faces; shear bars will be set perpendicular, not inclined.

g. Alternatives to the detailed design requirements noted in Items (a) through (f) shall be subject to the approval of the High Commission for Industrial Security.

22. A raised access, modular-type floor system, commonly referred to as a computer floor, shall be installed. If a computer room is to be incorporated into the control building, the raised access flooring should be installed in both areas. Care should be exercised to maintain fire code construction requirements and to provide adequate under-floor ventilation.

23. The ceiling system for control room building shall be suspended, exposed grid system with acoustical lay-in tiles and recessed lay-in lighting fixtures. A minimum clearance of 9 feet shall be provided between the finished floor and the ceiling system. The height of the control board and the relationship between the top of the board and the ceiling shall be considered.

Plumbing

24. Restroom facilities are required for control room personnel only. Quality standards shall be set forth in the design specifications.

25. Minimum installation consists of the following equipment and fixtures installed in a room accessible from, but not directly opening into, the control room:

a. Water closet enclosed in toilet compartment.

b. Lavatory, wash hand basin with soap dispenser, towel dispenser and mirror.

c. Water heater, either the electric instantaneous type for smaller installations, or the electric high recovery storage type, for larger installations.

d. A water service sink may be installed in the mechanical room, taking into consideration all electrical fitments installed.

e. An electric refrigerated drinking fountain should be installed convenient to the control room.

f. Suitable drainage for the toilet and wash hand basin systems and for the refrigeration and air-conditioning equipment shall be provided.

26. The control room building sewer shall be tied into the main plant sanitary sewer system in such a manner as to prevent the possibility of the back-up of explosive materials into the building.

27. The water supply system shall be connected to the main plant drinking water supply system/source.

All underground water piping shall be type 'K' copper tubing with wrought copper solder-joint fittings. Exposed water piping should be insulated as necessary for protection against local climatic conditions, this includes operational conditions.

28. Piping sleeves shall be installed where piping penetrates concrete foundations or exterior walls. These sleeves shall be standard-weight black steel pipe, one pipe size larger than the pipe passing through the sleeve. Annular space between the pipe and sleeve shall be caulked with flexible compound to prevent the passage of air, gas or rodents through the wall. Steel plate rings should be installed as escutcheon plates covering and protecting the annular space caulking from blast forces. Sleeves through concrete floor slabs shall be 26 gauge galvanized sheet metal, sized and caulked to prevent the passage of air.

Electrical

29. The design and installation of the control room electrical system shall be fully compatible with the main plant power and lighting systems. All electrical equipment should be located in the mechanical room, except where the magnitude of an installation may require that separate rooms be provided.

All conduits entering the building shall be underground.

30. Lighting requirements are as follows:

a. Recessed fluorescent lighting fixtures shall be installed for interior lighting. Ballasts for each fluorescent fixture shall be wired in series with a radio frequency suppressor.

b. Emergency lighting shall consist of a battery operated system.

c. Lighting fixtures and receptacles shall be compatible with the main plant electrical standards.

31. Conduit systems for intercom and telephones shall enter the building underground and shall be included in the design specifications.

32. Instrument power shall be a separate system, with the instrument power panel board and wiring completely independent of the electrical system supplying lighting and building equipment. This procedure prevents the possibility of stray signals being transmitted into sensitive instrument control devices. Instruments and instrumentation wiring shall be shielded within a 9 feet radius of a fluorescent light fixture or its electrical supply wiring.

33. Supplement power systems are required for lighting, building pressurization system and instrument power.

34. Instrument signal entry cables or tubing, or both, shall enter the control room at two or more separate locations, if possible. Plot plan considerations shall determine the number of entries if more than two. Multiple entries will minimize hazards associated with readily accessible common junction points. Signal runs within the control room should be designed to minimize common or bundled runs. Layout of the cross-connection panels should reflect these criteria.

35. When cables pass through fire walls and partitions, the wall openings shall be fully sealed to prevent transmission of fire and vapors. The opening shall be protected by a steel framed transit assembly with packing assembly for fire and watertight seal around the cable.

    All cables are to enter the building underground.

36. All cable entries shall be sealed. Multi-conductor cable covering shall be stripped and sealing shall be placed between the outside covering and the cables. A plotting compound shall be used to exclude the passage of hydrocarbon vapors and products of combustion into the control house via the cable covering.

    Special Mechanical Requirements

37. Self adjusting louvers in exterior walls shall be blast resistant.

38. Duct work supplying cooling to electronic instruments in the process control-boards shall be insulated from the control-board frame or structure by means of non-conductive, flexible connections.

39. Filter system design for air supply equipment is especially critical where computer installations are anticipated. The installations of electronic cleaners are recommended. Filter requirements shall be included in the design specifications.

40. Performance guarantees are required for each system. These shall maintain the following conditions throughout the control room on a 24-hour, year round basis.

    a. Internal space pressure equal to a minimum of 0.25 inches of water.

    b. Interior temperatures of 75º F dry bulb with a relative humidity of 40 - 50 per cent, with outside conditions based on recognized local design conditions.

c. A ventilation rate of 15 cfm per person or 1/30 of the volume of conditioned space, whichever is greater.

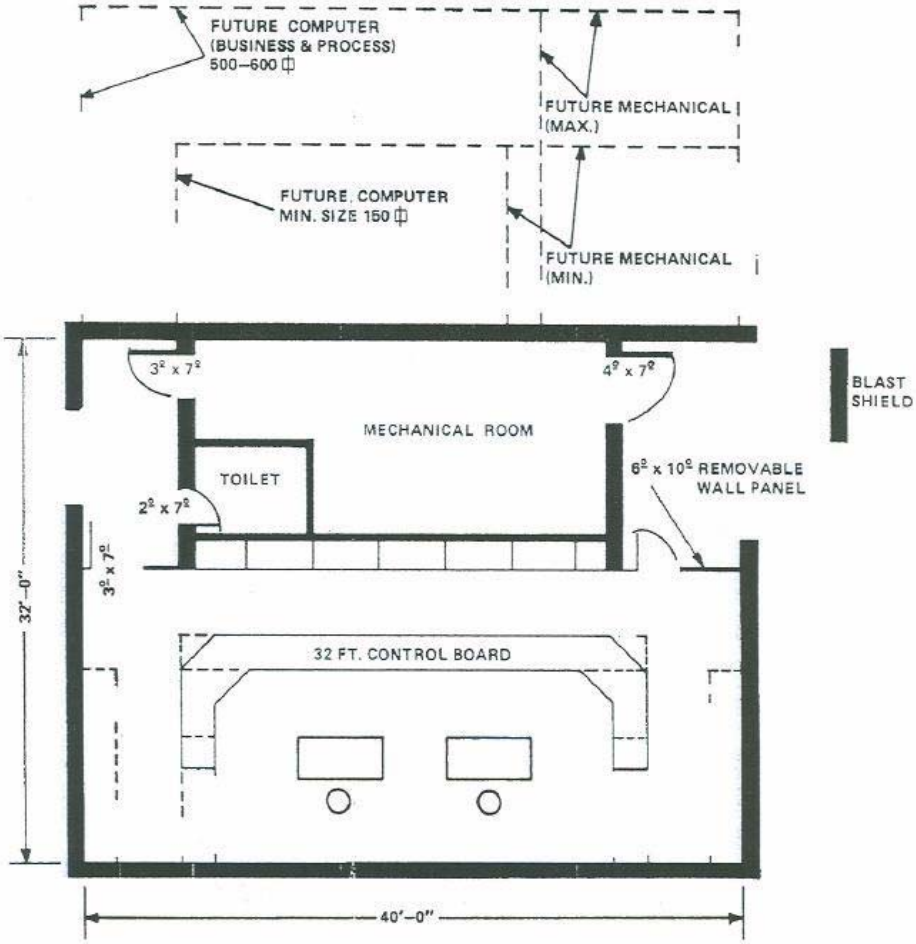
d. Full lighting, equipment and personnel loads. Changes and adjustments necessary to assure uniform conditions throughout the control room may be required during the first year of operation. A relative humidity instrument shall be installed in the control room.

41. All walls shall be painted white and all doors and frames shall be painted black.

Fire Protection

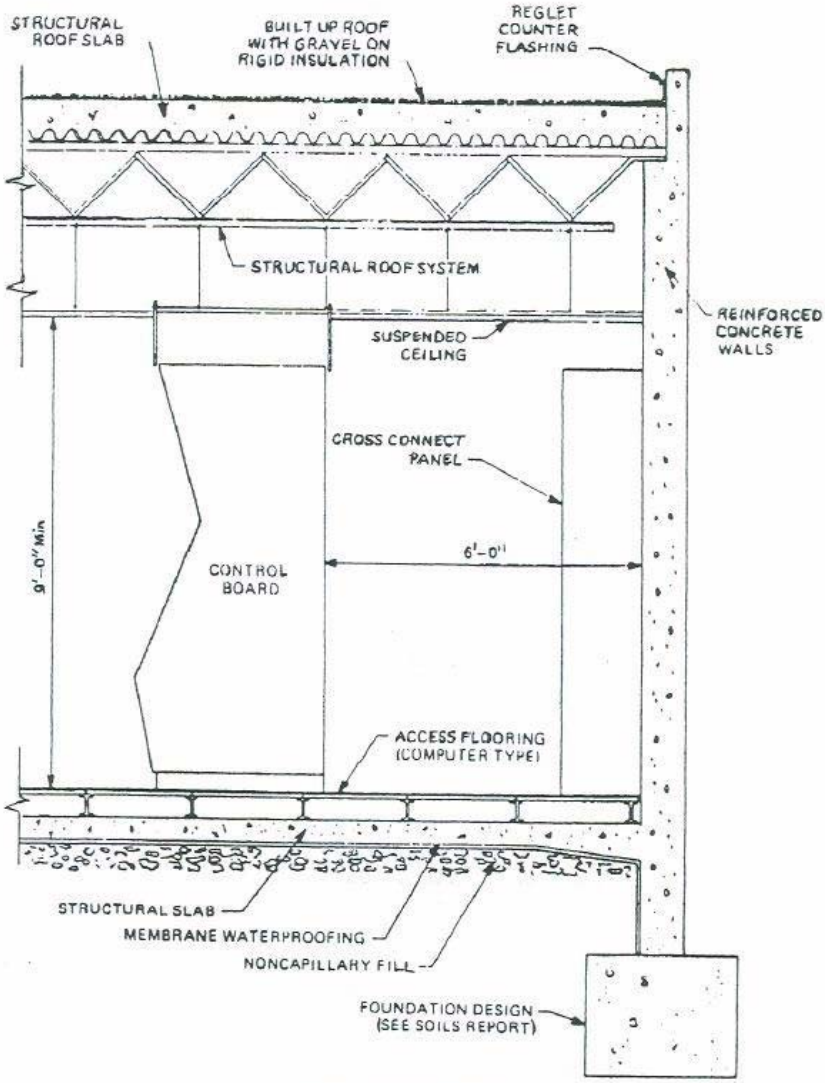
42. A fire alarm system for combined-function buildings will be installed where automatic or unattended equipment is housed adjacent to the control room. This is particularly recommended for rooms housing switchgear equipment. Automatic smoke detectors connected to the alarm system in the control room shall be installed in the switchgear room.

Unquote



NOTE: Size of control room depends upon the length and arrangement of the unit control boards.

CONTROL ROOM BUILDING
26 - 1A - 2



TYPICAL WALL SECTION

26 - 2A - 2

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
3					

TES-S-101.29

Revision No. 0

SAFETY & SECURITY – SPACING / DIKES AND DRAINAGE

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1. SCOPE

This Standard refers to the requirements for hydrocarbon storage tanks and is adopted verbatim from SAG SSD/27.

Quote

The following safety requirements are in addition to those normally found in internationally recognized Safety Codes and Practices, such as - NFPA 30, Mobile Engineering Guides and API Standards.

1. Spacing

a. Tanks of any diameter. Spacing between shells will be a minimum of one half the diameter of the largest tank.

b. Spacing between the tank shell and inside of the dike wall shall be a minimum of 15m.

c. Crude tanks in any diameter. Distance between tank shells will be a minimum of one and a half diameters of the largest tank.

d. Hot Oil tanks above 250º F and heated FCC Change Tanks, excluding High Flash Asphalt. Minimum spacing, shell to shell, will be equal to one and a half times the diameter of the largest tank.

e. Slop oil and Emulsion breaking tanks. Minimum spacing, shell to shell, will be equal to one and a half times the diameters of the largest tank.

f. All tanks, other than those which have to be individually diked and spheres, with built in common dikes shall not be located in more than two rows in any direction. There will not be more than four tanks built in a common dike to ensure easy access for fire fighting vehicles/equipment.

g. LPG spheres in any diameter. Distance between sphere shells will be a minimum of one and a half diameters of the largest sphere.

2. Dikes

a. Crude tanks in any diameter must be individually diked.

b. Hot oil, FCC Charge and Slop Tanks of any diameter, must be individually diked.

Note : All above dikes must contain at least 100% of the largest tank volume.

c. Tanks containing the same type of product, other than those listed in 2.1 and 2.2 above, may be located within a common dike. These to be limited to a maximum of four tanks.

d. Walls of all diked areas must be of solid masonry or concrete. Concrete slabs fitted over earthen dikes are permissible. All dikes must be designed to be liquid tight and to withstand a full hydrostatic head, with a minimum height of 2m from the interior grade level.

e. Intermediate dikes must not be more than 50% of the height of the main impounding area dike.

f. Impounding Dike Basins. A slope from the tank of not less than 1% shall be provided for at least 50 feet, or to the dike base, which ever is less.

g. Drains should extend through intermediate dikes, be of sufficient size and controlled with a drain valve.

h. The volumetric capacity of the diked area must not be less than the greatest amount of liquid that can be released from the largest tank within the diked area, assuming the largest tank is full.

i. To permit easy access for the fire fighting vehicles and/or equipment, there must be a minimum width of 6m around each impounding dike.

j. Only piping to/from individual tanks is permitted inside the diked areas. Non-associated pipe ways and all pumps must be located outside of the diked areas.

3. Drainage

a. Valved, fire sealed drains must be provided. They must be located at a remote part of the dike.

b. All valves, 14 inch and above, located on tank shell nozzles must be remotely operated to permit rapid emergency shut down.

c. The minimum size of the sealed drain pipe will be the diameter of the largest pipe to or from the tank.

d. Remotely operated main valves on tank nozzles, shall be protected against fire exposure by the application of fire proofing material to exposed conduit, motors, and valve bodies.

e. All storage tanks must be provided with shut-off valve below the liquid level. They must be located outside the dike wall.

f. Elevated LPG spheres must have the high point of the grade beneath the center of the sphere.

g. All bottom inlet lines on LPG spheres must have a remote operated valve located on the vessel nozzle to permit rapid shutdown. Each inlet/outlet line must be provided with an excess flow valve which would automatically shut off the flow in the event of a line rupture.

Unquote

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
3					

TES-S-101.30

Revision No. 0

SAFETY & SECURITY – THE SAFE DESIGN, CONSTRUCTION, OPERATION AND MAINTENANCE OF ON-SHORE PIPELINES FOR THE CONVEYANCE OF PETROLEUM (CRUDE OIL AND LIQUID PRODUCTS) AND GAS (NATURAL GAS AND GASEOUS PETROLEUM PRODUCTS)

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16.0 CLEARANCE FROM KNOWN HAZARDS

This standard is adopted verbatim from SAG SSD/29.

Quote

1.0 SCOPE

1.1 This Directive defines the minimum mandatory safety requirements for design, construction, operation and maintenance of all on-shore pipelines for the transportation of petroleum (crude oil and liquid products) and gas (natural gas and gaseous petroleum products). Such pipelines include, but are not limited to cross-country pipelines, flow lines, trunk lines, tie lines, gas lift lines and test lines. This Directive does not apply to in-plant piping, that is, piping within the physical boundaries of a processing facility.

1.2 The requirements of ASME B31.4 and ASME B31.8 are incorporated in this Directive by reference and shall apply to all pipelines within the scope of this Directive. The latest edition of ASME shall apply at all times.

2.0 PROJECT APPROVALS, CONFLICTS AND DEVIATIONS

2.1 All pipeline project proposals covered under this Directive shall be submitted to the High Commission for Industrial Security (HCIS) for approval during the initial planning stages of the project. No construction shall commence until approval has been granted. The proposal shall address all requirements of this Directive relating to the design, construction, operation and maintenance of the proposed pipeline. Subsequent relevant information related to the project shall be submitted to the HCIS regularly or on demand, as required**.**

2.2 Any conflicts between this Directive and other standards shall be resolved by the HCIS in consultation with the pipeline owner.

2.3 Any deviation from this Directive requires written approval from the HCIS.

3.0 PIPELINE LOCATION CLASSES

3.1 Pipeline location classes, by which ASME construction types are determined, shall be based on a population density analysis within the rupture exposure radius (RER) along the pipeline route.

3.2 For purposes of this directive, the RER is the downwind dispersion distance at ground level of a full bore rupture to the limit of 100 ppm H_2S or 1/2 the Lower Flammable Limit (LFL) of the released vapor, whichever distance is greater.

4.0 RUPTURE EXPOSURE RADII

4.1 Pipelines carrying liquid hydrocarbons having a true vapor pressure less than 14.5 psig and an H_2S concentration of less than 1.5 mole percent:

a. Shall have an RER distance of 200 meters (660 ft) for pipe diameters less than or equal to 24 inches.

b. Shall have an RER distance of 400 meters (1310 ft) for pipe diameters greater than 24 inches.

Table 1 - Pipeline RER Summary, Low TVP*

Pipeline Contents	Pipeline Diameter(in.)	H_2S Mole %	RER (m)
Sweet	≤ 24	< 1.5%	200
Low TVP & Liquid Only	> 24	< 1.5%	400

Note: *True Vapor Pressure < 100 kPa at 38°C

4.2 Pipelines carrying liquid hydrocarbons or combustible gas having a true vapor pressure of 14.5 psig or greater, and with an H_2S concentration of less than 1.5 mole percent:

a. Shall have an RER distance of 1000 meters (3280 ft) for pipe diameters less than or equal to 24 inches.

b. Shall have an RER distance of 2000 meters (6560 ft) for pipe diameters greater than 24 inches.

4.3 Pipelines carrying liquid hydrocarbons or combustible gas and with an H_2S concentration of 1.5 mole percent or greater:

a. Shall have an RER distance of 3000 meters (9840 ft) for pipe diameters less than or equal to 24 inches.

b. Shall have an RER distance of 5000 meters (16400 ft) for pipe diameters greater than 24 inches.

Table 2 - Pipeline RER Summary, Low TVP* or Gas

Pipeline Contents	Pipeline Diameter (in.)	H_2S Mole %	RER (m)
Liquid or Gas	≤ 24	< 1.5 %	1000
	≤ 24	≥ 1.5 %	3000
	> 24	< 1.5 %	2000
	> 24	≥ 1.5 %	5000

Note: *True Vapor Pressure ≥ 100 kPa at 38°C

4.4 The RER for a flow line shall be equal to the RER of the well served. For gathering lines the RER shall be equal to the largest RER of any well gathered by that line.

4.5 An RER calculation may be used to reduce the RER provided the calculation is based upon the pipeline diameter, fluid composition, temperature and maximum shut-in pressure. The results shall be approved by the Chief Engineer of the pipeline owner and the HCIS.

4.6 Regardless of calculations, RERs shall not be less than the values specified in section 4.1 nor greater than the values specified in sections 4.2 and 4.3 for the respective pipeline diameters.

5.0 POPULATION DENSITY ANALYSIS

The following paragraphs supersede the requirements in ASME B31.4 and ASME B31.8 pertaining to the population density index:

5.1 The boundaries of areas in which development is present or planned within the RER of the pipeline shall be indicated by the pipeline owner.

5.2 The population density index for a pipeline is the sum of the existing density index and virtual density index values for each section of the line and shall be used as the design basis of the line.

5.3 Buildings having more than four occupied stories shall be included in the density index as a number of equivalent buildings. The number of equivalent buildings shall be calculated by dividing the number of stories in those buildings by there (3) and rounding up the whole number.

5.4 The existing density index for a location shall be determined from a count of the number of buildings lying within the RER of the pipeline.

a. An existing density index shall be calculated for each specific 1 kilometer (0.6 mile) section of pipeline.

b. To determine the existing density index for a pipeline, establish a zone one RER wide to each side of the pipeline. Divide the pipeline and associated RER zone into 1 kilometer (0.6 mile) long sections. Count the number of buildings and equivalent buildings in each of the sections. This whole number count is the existing 1 kilometer density index for the section.

5.5 In areas where development is planned, the estimated number and function of future buildings are used to determine the virtual density index.

a. A virtual density index shall be calculated for each specific 1 kilometer (0.6 mile) of pipeline.

b. To determine the virtual density index of a pipeline, establish a zone one (1) RER wide to each side of the pipeline route. Divide the pipeline and associated RER zone into 1 kilometer (0.6 mile) long sections.

c. Calculate the land area in square meters of any planned development which is included within the zone.

d. Multiply the included area by 0.00075 (exactly) and round up. The resulting whole number is the virtual density index for this section.

5.6 Temporary facilities which will be in place for less than six (6) consecutive months shall not be included in these calculations.

5.7 The extent of RER zones, the boundaries between location class areas and the location class designation shall be marked on plan drawings. Additionally the population density index for each 1 kilometer (0.6 mile) of pipeline shall be provided on a listing in all pipeline project proposals.

6.0 PIPELINE LOCATION CLASSIFICATION

6.1 Class 1: Class 1 locations are undeveloped areas for which the population density index* for any 1 kilometer (0.6 mile) section is ten (10) or less.

Class 2: Class 2 locations are areas for which the population density index* is eleven (11) through thirty (30) or which include primary or secondary highways as defined by the Ministry of Communications.

Class 3: Class 3 locations are areas for which the population density index* is more than thirty (30).

Class 4: Class 4 locations are areas in which a school, hospital, hotel, prison, shopping mall or similar retail complex is located, as well as Class 3 areas which include buildings or more than four (4) occupied floors.

*Note: For population density index refer to section 5.2

6.2 No pipeline having a diameter of greater than 24 inches shall be permitted within 1000 meters (3280 ft) of any hospital, school or university.

6.3 Drainage for pipelines located within 400 meters (1300 ft) of residential areas and with high population densities (Class 3 and 4 locations) shall be designed to prevent liquids from traveling toward those areas.

6.4 Continued surveillance by the pipeline owner should be maintained for existing and completed pipelines to determine if additional buildings, intended for human occupancy, have been constructed within a Location Class.

7.0 DESIGN PRESSURE

7.1 Design shall meet the standards specified in ASME B31.4 and ASME B31.8.

7.2 The design pressure used to determine the minimum pipe wall thickness as required by ASME B31.4 and ASME B31.8 in location Class 3 and Class 4 zones shall be established by determining the maximum expected surge pressure from a single contingency, such as inadvertent closure of a valve or failure of a sensing or regulating device. Self activated surge protection systems, if provided, shall be assumed to operate as intended, to mitigate the single worst contingency.

7.3 Surge analysis shall be made for liquid-packed services. Surge protection systems shall be installed if surge pressures are calculated to exceed 110% of the maximum allowable operating pressure.

7.3.1 Surge protection systems shall include duplicate critical sub-assemblies sufficient to overcome single-mode system failures.

7.3.2 An installed spare surge relief valve is required for each surge protection system.

7.3.3 Surge protection systems shall be of fail-safe design.

8.0 SCRAPER LAUNCHER AND RECEIVER INSTALLATIONS

8.1 All pipelines passing through location Class 3 or Class 4 zones shall have appropriate connections for either permanent or temporary scraper launchers and receivers.

Within producing areas scraper connections are not required on flow lines, trunk lines, test lines and dry gas lift lines.

9.0 PIPELINE CORRIDORS

9.1 All pipelines shall be placed in designated corridors, which are a minimum of 60 meters (200 ft) wide. A pipeline corridor constitutes an exclusive land use area for pipeline related activities; no other uses are allowed, except as specifically addressed herein.

Within producing areas, flow lines, trunk lines, test lines and dry gas lift lines with diameters of 24 inches or less shall have a minimum corridor width of 6.5 meters (21 ft).

9.1.1 Where the pipeline will cross the right of way (ROW) belonging to others, approval shall be obtained from the HCIS and concerned authorities before any development takes place.

9.1.2 With approval of the HCIS and the pipeline owner, an owner's pipeline corridor may be used for nonowner fiber optic or other communication cables placed parallel to the pipeline. Pipeline corridors may be used for fiber optic or other communication cables placed parallel to the pipeline if used solely for the pipeline owner's operational purposes.

9.2 The minimum clear space between any pipeline and the corridor boundary shall be 30 meters (100 ft).

Within producing areas, for flow lines, trunk lines, test lines and dry gas lift lines with diameters of 24 inches or less, the minimum clear space between any pipeline and the corridor boundary shall be 1 meter (3.3 ft).

9.3 Spacing of parallel buried pipelines shall be as follows:

9.3.1 Every pipeline must have a minimum clearance of 15 meters (50 ft) on one side.

9.3.2 Within producing areas, flow lines, trunk lines, test lines and dry gas lift lines with diameters 24 inches or less must have a minimum clearance of 4.5 meters (15 ft) on one side for maintenance vehicle access. Within 450 meters (1500 ft) of a plant fence, this minimum clearance is not required, nor is it required at pipeline crossings, tie-ins, or for short sections less than 20 meters (66 ft) in length.

9.3.3 Two pipelines may be located near one another provided that a minimum clearance equal to the greater of the following is maintained between them:

a . One meter (3.3 ft) or,

b. Twice the diameter of the larger pipe.

9.3.4 The corridor shall be graded and/or bermed to drain any spillage from a rupture of the pipeline away from the pipeline itself, roads, bridges, flyovers, or any other adjacent structure.

9.4 Where the pipeline and power line corridors cross, the horizontal angle of intersection shall be within forty-five (45) to one hundred and thirty five (135) angular degrees for power lines operating at 69 kV and above.

9.4.1 In case the angles in 9.4 cannot be met, an analysis of normal and transient voltages shall be made and the requirements of paragraph 9.4.3 shall be met.

9.4.2 The minimum horizontal separation between any pipeline and the poles or structure of overhead power lines shall be 30 meters (100 Ft) for power lines operating at 69 kV and above.

9.4.3 The following minimum spacings between a pipeline and a parallel overhead power line shall be maintained ('up to' means 'up to and including').

For any voltage, greater than 69 kV, parallel for more than 1.6 km (1 mile)	150m
Up to 230 kV, parallel for less than 1.6 km (1 mile)	30m
Up to 500 kV, parallel for less than 1.6 km (1 mile)	50m
Up to 1000 kV, parallel for less than 1.6 km (1 mile)	85m

9.4.3.1. Induced voltages in pipelines and communication cables shall not exceed twelve volts (12 V) under normal steady state conditions. Under fault conditions, induced voltages shall not exceed the values calculated for formulas in ASME/IEEE Standard 80. The need for an analysis of normal and transient voltages shall be determined by the pipeline owner.

9.4.4 The minimum vertical clearance, measured from grade of overhead power line conductors at pipeline crossings shall be 9 meters (30 ft) for power lines operating at 230 kV or less and 13.5 meters (45 ft) for power lines operating at over 230 kV.

9.5 Underground cables crossing the pipeline corridors shall do so by means of a concrete duct bank. The minimum vertical distance between the bottom of any pipe and the top of the duct banks shall be 1 meter (3.3 ft). The concrete duct bank shall be continuous, and at the same elevation across the full width of the corridor. There shall be no servicing points for cables within the pipeline corridor except where the cables belong to the pipeline owner, when cable servicing points shall be no closer than 25 meters (82 ft) to any pipeline in the corridor.

9.6 Piping crossing pipeline corridors shall cross pipelines in a manner such that the vertical distance between pipeline surfaces is at least 0.5 meters (1.5 ft). Unless belonging to the owner of the corridor, the piping shall have no servicing points within the pipeline corridor.

9.7 Pipelines shall not be laid under any road/expressway except for crossing. The crossing should be at right angles to the road/expressway, but shall not be at an angle of less than 45°. Where a new pipeline crosses an existing road/expressway, it shall be buried directly in the ground or it may be cased. The diameter of the casing shall not be less than the outside diameter of the pipe plus four inches (4"). The minimum depth of cover of a pipeline at the road/expressway shall be 1.2 meters (4 ft). Reduction in this depth may, in consultation with the HCIS, be acceptable, provided it is supported by calculations of load stress based on type of soil and frequency and type of traffic. Crossings shall be designed for the highest total loading expected. Records shall be kept of these design calculations and shall be made available to the HCIS on demand. If it is intended to follow the route of a road/expressway. The pipeline shall not be less than 100 meters (330 ft) from the center of the road/expressway.

9.8 Where a new road crosses an existing pipeline, the road shall be carried by a flyover. Alternative crossings, such as by casing, by culvert, or by direct burial, may be acceptable following consultation with the HCIS. Access and egress for operation, maintenance and inspection vehicles shall be provided to the pipeline right of way (ROW) within 1000 meters (3280 ft) of the crossing.

9.8.1 Should culverts be used, the minimum spacing between any pipe and the overhead or side portions of the culvert should be 1.5 times the pipe diameters, but in no case shall it be less than 1 meter (3.3 ft).

9.9 At road / expressway crossings the pipeline shall be protected against vehicle collision and damage by means of suitable traffic barriers.

9.10 Crossings shall be designed for the highest total loading expected. Railway loadings shall be based on three (3) cross ties, each carrying sixty five (65) metric tons (seventy two (72) short tons).

9.11 Casings shall be used for all railroad crossings.

10.0 PIPELINE VALVES

10.1 Sectionalizing block valves and emergency isolation valves shall be installed in accordance with ASME B31.4 and ASME B31.8

10.2 Emergency isolation valves shall be installed between pipeline sections of:

location Class 1 and Class 3,
location Class 1 and Class 4,
location Class 2 and Class 3,
location Class 2 and Class 4, and
location Class 3 and Class 4.

Exception: In producing areas for flow lines, trunk lines, test lines, and dry gas lift lines with diameters of 24 inches or less, emergency isolation valves are not required.

10.3 All required emergency isolation valves shall be installed within the lower location Class at least the RER distance beyond the boundary between the Class areas.

10.4 The extension of a pipeline section of higher strength ASME location Class design into a lower location Class area beyond the required RER distance is permissible to minimize the number of emergency isolation valves.

10.5 Adjustments in pipeline wall thickness shall comply with ASME B31.4 and B31.8. Where an emergency isolation valve is required, a change in ASME location Class design can only be made at the emergency isolation valve.

10.6 The emergency isolation valves for location Class 3 and Class 4 areas shall be fitted for remote pressure monitoring and closure activation capability from a manned control facility.

10.7 The emergency isolation valve closure rate shall be determined by the pipeline's surge characteristics, but total closing time shall not exceed ten (10) minutes.

11.0 BURIAL OF PIPELINES DURING INSTALLATION

11.1 In general, all major cross-country pipelines carrying crude oil, gas or hydrocarbon products shall be buried and the original ground surface restored or graded. Pipelines shall be laid in virgin ground with a minimum depth of cover of 2 meters (6.6 ft). If there are engineering or economic considerations which justify exceptions to these requirements, such as unstable or rocky soil conditions or corrosive environments, alternative installation methods may be considered with less cover if additional protection is provided to withstand anticipated external load, and to minimize damage to the pipe by external forces; however, all variances must first be approved by the HCIS.

11.2 The pipeline shall be laid in a bed of sand or other fine grain material. The bed shall be 1.5 times the nominal diameter of the pipeline. Adequate pipeline location markers, indicating caution for the protection of the pipeline, the public and persons performing work in the area, shall be installed over each line on each side of the road, highway, railroad and stream crossings, and the frequency of the location markers should be increased in areas of high population density.

11.3 Backfill shall be free of material that may damage coatings of steel casings or carrier pipe. Backfill shall be placed in layers 0.3 meter (1 ft) or less.

11.4 At road crossings, the road pavement and sub-base shall be restored. The surface shall be compacted and finished flush with the adjoining pavement.

12.0 TESTING AND INSPECTION

12.1 All carrier pipes shall be visually inspected prior to installation.

12.2 Welds in carrier pipes shall be inspected by nondestructive test methods prior to installation.

12.3 Throughout installation, the welding quality should be checked in accordance with ASME B31.4 and ASME B31.8.

13.0 CORROSION PROTECTION

13.1 The pipeline shall be protected from corrosion by the application of an appropriate external coating system such as fusion bonded epoxy and by cathodic protection.

13.1.1 The voltage on the pipeline shall be continuously monitored for potential and polarity.

13.2 Casings welded to carrier pipes shall be considered part of the pipeline.

13.3 A separate system for cathodic protection shall be used for casings insulated from the carrier line.

14.0 PIPELINE OPERATION

14.1 The pipeline shall not be operated outside the criteria used in its design, and shall be maintained at all times in a fully operational condition.

14.2 The pipeline shall be inspected regularly for the effects of corrosion and erosion. When the design limits are reached, the affected portion of the pipeline shall be taken out of service. Records of maintenance and replacement items shall be kept available for inspection by the HCIS on demand.

15.0 ABANDONED PIPELINES

15.1 In the event of abandoning a pipeline, it is required that:

15.1.1 Facilities to be abandoned in place shall be disconnected from all sources of liquid or gas, such as other pipelines, crossover piping, meter stations and control lines.

15.1.2 Facilities to be abandoned in place shall be purged of the transport liquid or gas with an inert material.

16.0 CLEARANCE FROM KNOWN HAZARDS

16.1 With the exception of producing lines, pipelines shall not be closer to the following hazardous locations than the distances given below.

16.1.1	Oil wells, gas wells and gas/oil separation plants (GOSPs), except for termination at the GOSP or well	100m
16.1.2	Pressure crude pipelines (minimum spacing between any two pipelines should be twice the diameter of the larger pipe, but in no case less than 1 m). (One side only).	15m
16.1.3	Natural gas or LPG pipelines (minimum spacing between any two pipelines should be twice the diameter of the larger pipe, but in no case less than 1 m). (One side only).	15m
16.1.4	Atmospheric crude flow lines a. below ground b. exception, 9.3.2 within producing areas	 15m 4.5m
16.1.5	Buried pipeline and ground flare or burn pit	15m
16.1.6	Elevated flare stack, over 50 ft.	60m
16.1.7	Oil/water separators	45m
16.1.8	Hydrocarbon processing areas	45m
16.1.9	Explosive and capsule storage areas	5000m
16.1.10	Gas bottling and storage areas	1500m
16.1.11	Airfield, except for termination at the airfield	450m
16.1.12	Radio, television, and microwave towers in support of pipeline operations. (Towers unrelated to pipelines to be located outside the pipeline corridor)	15m
16.1.13	Main railway or spur line	60m
16.1.14	Sewer	10m
16.1.15	Underground power cable (except where crossing the pipeline).	4m
16.1.16	Underground liquid hydrocarbon storage tank	35m